

AN EXPLAINABLE AND ACCESSIBLE MACHINE LEARNING FRAME- WORK FOR OVERACTIVE BLADDER SEVERITY CLASSIFICATION USING MACHINE LEARNING ALGORITHMS.

*Report submitted to the SASTRA Deemed to be University as the
requirement for the course*

CSE300 - MINI PROJECT

Submitted by

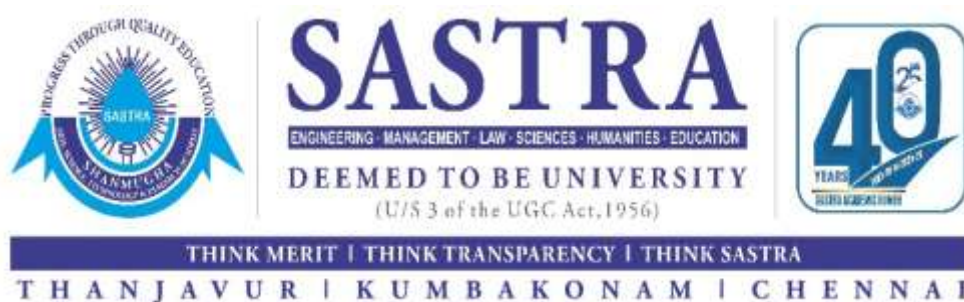
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Bonafide Certificate

This is to certify that the report titled” **An Explainable and Accessible Machine Learning Framework for Overactive Bladder Severity Classification Using Hybrid Algorithms**” submitted as a requirement for the course, CSE300 : **MINI PROJECT** for B. Tech is a Bonafide record of the work done by Ms. Arunthathi S (Reg. No. 227003017, B. Tech CSE), Ms. Jaishree A (Reg. No. 227003059, B. Tech CSE) and Ms. Padmapriya K (Reg. No. 227003101, B. Tech CSE) during the academic year 2025-26, in the School of Computing, under my supervision.

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Abbreviations

AI	Artificial Intelligence
ANN	Artificial Neural Networks
ANOVA	Analysis of Variance
ATP	Adenosine Triphosphate
AUC	Area Under the Curve
BDNF	Brain Derived Neurotrophic Factor
BPS	Bladder Pain Syndrome
CAT	Context Aware Thresholding
CNN	Convolutional Neural Network
COS	Core Outcome Set
COVID-19	Coronavirus Disease 2019
F1	F1 Score
FN	False Negatives
FP	False Positives
FPR	False Positive Rate
HF	Hugging Face
IC	Interstitial Cystitis
IC-PIS	Interstitial Cystitis Personalized Inflammation Score
ICIQ	International Consultation on Incontinence Questionnaire
ICPI	Interstitial Cystitis Problem Index
ICSI	Interstitial Cystitis Symptom Index
LASSO	Least Absolute Shrinkage and Selection Operator
LDA	Linear Discriminant Analysis
LE8	Life Essential Eight
LIME	Local Interpretable Model Agnostic Explanations
LR	Logistic Regression
MCC	Matthews Correlation Coefficient
ML	Machine Learning
MLP	Multi-Layer Perceptron

NGB	Neurogenic Bladder
NGF	Nerve Growth Factor
OAB	Overactive Bladder
OABq	Overactive Bladder Questionnaire
OABSS	Overactive Bladder Symptom Score
PGE2	Prostaglandin E2
PGI-I	Patient Global Impression of Improvement
PFMT	Pelvic Floor Muscle Training
PPV	Positive Predictive Value
PRO	Patient Reported Outcomes
PUF	Pelvic Pain and Urgency Frequency
RCI	Relative Classifier Information
RF	Random Forest
ROC	Receiver Operating Characteristic
RUSBoost	Random Under Sampling Boosted
SDT	Static Detrusor Thresholding
SHAP	SHapley Additive Explanations
SMOTE	Synthetic Minority Oversampling Technique
SPSS	Statistical Package for Social Sciences
SVM+LR	Hybrid Support Vector Machine and Logistic Regression
SVM	Support Vector Machine
TN	True Negatives
TP	True Positives
TPR	True Positive Rate
UUI	Urgent Urinary Incontinence
VGG16	Visual Geometry Group 16
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

Notations

AUC	Area Under the Curve
F1	F1 Score
FN	Number of False Negatives
FP	Number of False Positives
FPR	False Positive Rate
K	Number of Folds in Cross Validation
N	Total Number of Classes
MCC	Matthews Correlation Coefficient Value
P	Precision Value
R	Recall Value
TN	Number of True Negatives
TP	Number of True Positives
TPR	True Positive Rate
X1	OAB Experience after COVID-19
X2	Daytime Urination Frequency
X3	Nighttime Urination Frequency
X4	Amount of Urine Leakage
X5	Pain or Discomfort in Lower Abdomen
X6	Difficulty in Urinating
X61	Urine Leakage Before Reaching Toilet
X7	Urination Frequency from Morning to Night
X8	Urination Frequency from Night to Morning

Greek Symbols

α	Significance Level in Statistical Testing
μ	Mean Value
σ	Standard Deviation
Σ	Summation

Miscellaneous Symbols

%	Percentage
<	Less Than
>	Greater Than
$\pm$	Plus or Minus
∞	Infinity
p	Probability Value in Hypothesis Testing

AN EXPLAINABLE AND ACCESSIBLE MACHINE LEARNING FRAMEWORK FOR OVERACTIVE BLADDER SEVERITY CLASSIFICATION USING MACHINE LEARNING ALGORITHMS

Abstract

Overactive bladder (OAB) is a prevalent urinary disease, where a person experiences a sudden, overwhelming urge to urinate, often leading to leaking of urine before making it to the toilet. OAB sufferers also experience frequent urination throughout the day and night (nocturia). This urinary condition has a detrimental effect on a person's daily routines, social activities and mental health, and has been shown to be more common in women than men. This is often difficult to detect in the early stages due to expensive, time-consuming and unavailable (in rural areas) diagnostic tools such as urodynamic testing. This research offers a machine learning approach to automatically determine the severity of OAB (No OAB, Mild OAB, and Moderate OAB) based on the responses from a questionnaire. Data was from 53 women (aged 17 to 53 years) through an online form. It included information on the number of day and night-time urinations, volume of unintentional urine loss, lower pelvic pain, urination difficulty, and OAB symptoms after COVID-19 infection. To enhance accuracy, features without statistical significance were eliminated using Analysis of Variance (ANOVA). Several machine learning models were developed and tested: Support Vector Machine (SVM), Artificial Neural Network (ANN), Linear Discriminant Analysis (LDA) and Random Under-Sampling Boosted classifier (RUS Boost). The original data set was low and imbalanced, so the Synthetic Minority Oversampling Technique (SMOTE) was used to balance and increase the data set from 53 to 900 patient samples. The models were tested with 5-fold and 10-fold cross-validation. The most accurate model was SVM with an accuracy of 97.78% (sensitivity 96% and specificity 97%) after using SMOTE. In order to facilitate trust and interpretation of the model for clinical practice, Explainable AI was applied. ShaPley Additive exPlanations (SHAP) was used to find the most important features used to make predictions across all patients, and Local Interpretable Model-Agnostic Explanations (LIME) was used to explain the prediction of each individual patient's OAB severity. Both approaches found trouble urinating, OAB after COVID-19, and leaking urine before reaching the toilet to be the most important features in predicting OAB severity. Finally, A publicly available web app on Hugging Face, enabling anyone to check their OAB severity at the tip of their fingers through a smartphone or PC, without having to go to a hospital or pay a cent.

KEY WORDS: *Overactive Bladder, Support Vector Machine, SMOTE, Explainable Artificial Intelligence,*

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1. SUMMARY OF THE BASEPAPER

Title: Predicting Overactive Bladder Symptom Severity with Explainable AI: A Data-Driven Approach Using Questionnaire Responses

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SUMMARY:

Over Active Bladder (OAB) is a frequent urologic condition, particularly in women, that results in an urgent and hard to suppress need to urinate. It has been typically diagnosed by expensive urodynamic tests and repeated visits to the doctor, which can be cumbersome for patients. In this work, Data was gathered about the patients in a quicker and cheaper way. Then these questions used to train ML algorithms to identify patterns and classify patients into three groups - No OAB, Mild OAB and Moderate OAB - to simplify and speed up the diagnosis.

1.1INTRODUCTION

OAB is overactive bladder. Patient with OAB face difficulty in day-to-day activities. Person feels embarrassed and stressed all the time. This condition affects millions of people worldwide [3]. Most people do not talk about it openly so they do not get treatment on time. Women are affected more than men [4]. In Asia the problem is even more common [4]. OAB affects around 11 percent of people globally [3].

Doctor has no test for this problem. Doctor only asks questions and gives opinion [6]. Two doctors give different answers for same patient. So, this is not reliable. Doctor appointment takes lot of time too. Poor people cannot afford costly tests. People in villages have no specialist doctors nearby. So, most people just suffer without knowing what exactly is wrong. Questionnaires like OABSS are used to measure symptoms [6]. But manual reading of these forms takes too much time [7].

Machine learning is now used in many medical fields [10]. It helps doctors make better decisions faster [11]. Big data and machine learning together can improve healthcare a lot [12]. Machine learning finds hidden patterns in data that humans cannot find easily [13]. In urology

also machine learning is now being used widely [14]. Various studies have shown that computer programs can predict urinary problems with good accuracy [15]. Artificial intelligence is changing the way doctors diagnose diseases [16].

For OAB specifically machine learning has shown very good results. SVM model was used to classify bladder conditions and gave good accuracy [9]. Real time classification of bladder events was done using machine learning [9]. Urinary tract infections were detected using AI with 95 percent sensitivity [8]. Cystitis was detected in ultrasound images using deep learning [19]. Bladder cancer was predicted using clinical lab data and machine learning [17]. Inflammatory process in bladder cystitis was analysed using machine learning [18]. Urinary cell free DNA was used to detect bacterial cystitis using spot-based machine learning [20]. Interstitial cystitis was classified using machine learning based predictions [21]. Urinary tract infections were predicted and classified using machine learning [22].

For OAB prediction specifically SVM gave very good results [23]. Machine learning was also used to predict response to OAB medications [24]. But most of these models are black box models. Doctors cannot understand why the model gave a particular result. So they do not trust these models easily.

Explainable AI solves this problem [25]. LIME explains predictions of any classifier in simple way [26]. SHAP values show how each feature contributes to the final prediction [27]. Making AI interpretable helps doctors trust and use these systems better [28].

Another problem in OAB datasets is class imbalance. Some groups have very few patients. This makes the model biased. SMOTE technique fixes this problem by creating extra data for smaller groups [29]. Learning from imbalanced data is now a well studied problem in machine learning [30].

This study just asked simple questions to female patients. Questions like toilet visits per day, whether urine leaks before reaching toilet, whether they wake up at night, whether stomach pain exists and whether urinating feels difficult. All answers came from 53 female participants through a simple Google form [6]. Ages were between 17 and 53 years. Based on answers patients were put into three groups. No OAB, mild OAB and moderate OAB. Four computer programs just checked all these answers. These are SVM, ANN, LDA and RUSBoost [39] [40]. SVM gave best result. Accuracy was 93 percent. After SMOTE was used accuracy went up to 96 percent [29]. SHAP values just showed which questions affected result most [27].

Difficulty in urinating and leaking before toilet came out as biggest factors. This helps doctors see why that answer came so they trust it better [25]. Apart from Base Paper, This project added a new thing. A website was put up on Hugging Face. Anyone can open it on phone or laptop. Answer few questions. Get result in seconds. No hospital visit needed. No doctor appointment needed. Very useful for people in villages. This whole thing is faster and cheaper. No costly machines needed. No painful tests needed. Just simple questions and computer programs are enough [8].

1.2 PROBLEM STATEMENT

OAB is hard to find. No proper test exists. Doctors just guess. Different doctors give different answers. Questionnaire based diagnosis takes lot of time. Manual checking by doctor is slow. Small dataset makes it harder. Class imbalance is also a problem. Existing methods are costly. Rural areas have no specialist doctors. So automatic detection is needed. Machine learning can help here. But black box models are not trusted by doctors. So explainable AI is also needed.

1.3 LITERATURE SURVEY

Our body has a small bag inside. This bag holds pee. We call it the bladder. Most people can wait before going to the toilet. They feel fine while waiting. But some people suddenly feel they must run to the toilet very fast. This happens even when the bag is not full. Doctors call this problem Overactive Bladder. They also call it OAB. Scientists have been studying this problem for almost twenty years. The first big challenge was that doctors from different countries were using different names for the same condition. Because of this they could not properly share their findings with each other. So, doctors from around the world agreed to use one common name for this problem [27].

After that a large study found that about one out of every nine people in the world have OAB. Older people and women tend to get it more than anyone else [1]. Women living in Asian countries said the worst part of OAB was that sudden scary urge to rush to the toilet without any warning [3]. A separate study went through twenty years of research and found that more people in Asia have OAB than in any other part of the world [2].

Doctors then needed a simple way to check how bad a patient's OAB was. So, they created a short list of questions to ask patients [7]. The questions are very easy. First question asks how many times a person goes to the toilet in a day. Second question asks how many times a person wakes up at night to use the toilet. Third question asks if any pee comes out before the person reaches the toilet. Each answer is given a number and all the numbers are added together to give one final score. This score tells the doctor how serious the condition is. Researchers tested these questions on many patients and found the answers were always steady and correct. That is why doctors all over the world still use these questions today [28].

After some time, computers began helping doctors with bladder problems. A computer program looks at thousands of patient records and learns from them. When a new patient comes in the program uses what it learned to figure out what might be wrong. One study showed that a computer program correctly found bladder infections in 95 out of every 100 patients and also made things easier for doctors and nurses [8]. Another team placed small sensors on the body and used a computer to watch the bladder all the time and that program gave the correct answer 97 times out of every 100 [9]. Scientists said these programs are very good at finding patterns

inside large amounts of medical information that doctors might not easily notice [19]. In some cases, these programs worked just as well as expert doctors when looking at medical pictures

[20]. A simple guide was also written so that more doctors could learn how to use these programs [21]. These programs also helped with other bladder problems. One program detected bladder cancer just by reading routine lab results [10]. Another program told apart different types of bladder swelling [11]. One program studied sound wave pictures of the bladder and spotted infections on its own [12]. One team found a way to detect bacterial infections by examining tiny pieces of genetic material found in pee and no painful test was needed for this [13]. Other teams used these programs for more difficult bladder conditions and kept getting strong results [14,15].

When researchers reviewed all these studies, they found that two types of programs called support vector machines and random forests worked better than all other types for bladder problems [16,17]. For OAB specifically one team built a program to find out how much pee was left in the bladder after a person used the toilet. This tells doctors how well the bladder is doing its job. That program scored 0.92 out of 1.00 which is considered a very strong result [4]. Another team created a program to predict whether a patient would feel better after taking OAB medicine. That program got the right answer 80 to 81 times out of every 100 and it worked well even when tested on completely new patients it had never seen before [6].

Even when a program works very well there is still one major problem. Doctors need to understand why the program gave a certain answer. Without knowing the reason, they cannot fully trust or use that answer. It is like a friend telling you to do something without explaining why. One team solved this using a method called SHAP. This method shows the doctor which information the program used the most and how much each piece of information changed the final answer. When doctors could see this clear explanation, they felt much more comfortable using the program [5]. SHAP gives each piece of information a score showing how much it influenced the result [23]. Another tool called LIME gives a simple explanation for just one answer at a time rather than explaining how the whole program works [22].

Scientists said this kind of explanation must be treated as a basic requirement in all medical programs because a doctor who does not understand why a program gave an answer simply cannot use it safely [24]. There is one more problem that comes up when building these programs. Some health conditions happen very rarely while others happen very often. This confuses the program because it keeps giving the common answer and ignores the rare one. It is like a student who studies only one topic before an exam and leaves everything else aside. To fix this a method called SMOTE was developed. It creates extra examples of the rare condition and adds them to the data so the program learns from a fair and balanced set [25]. When researchers compared many different ways to fix this problem SMOTE came out as the best one [26]. Lastly scientists reminded all that the safety of patient personal information should be considered at the very start before anything is initiated [18].

1.4 ARCHITECTURAL DIAGRAM

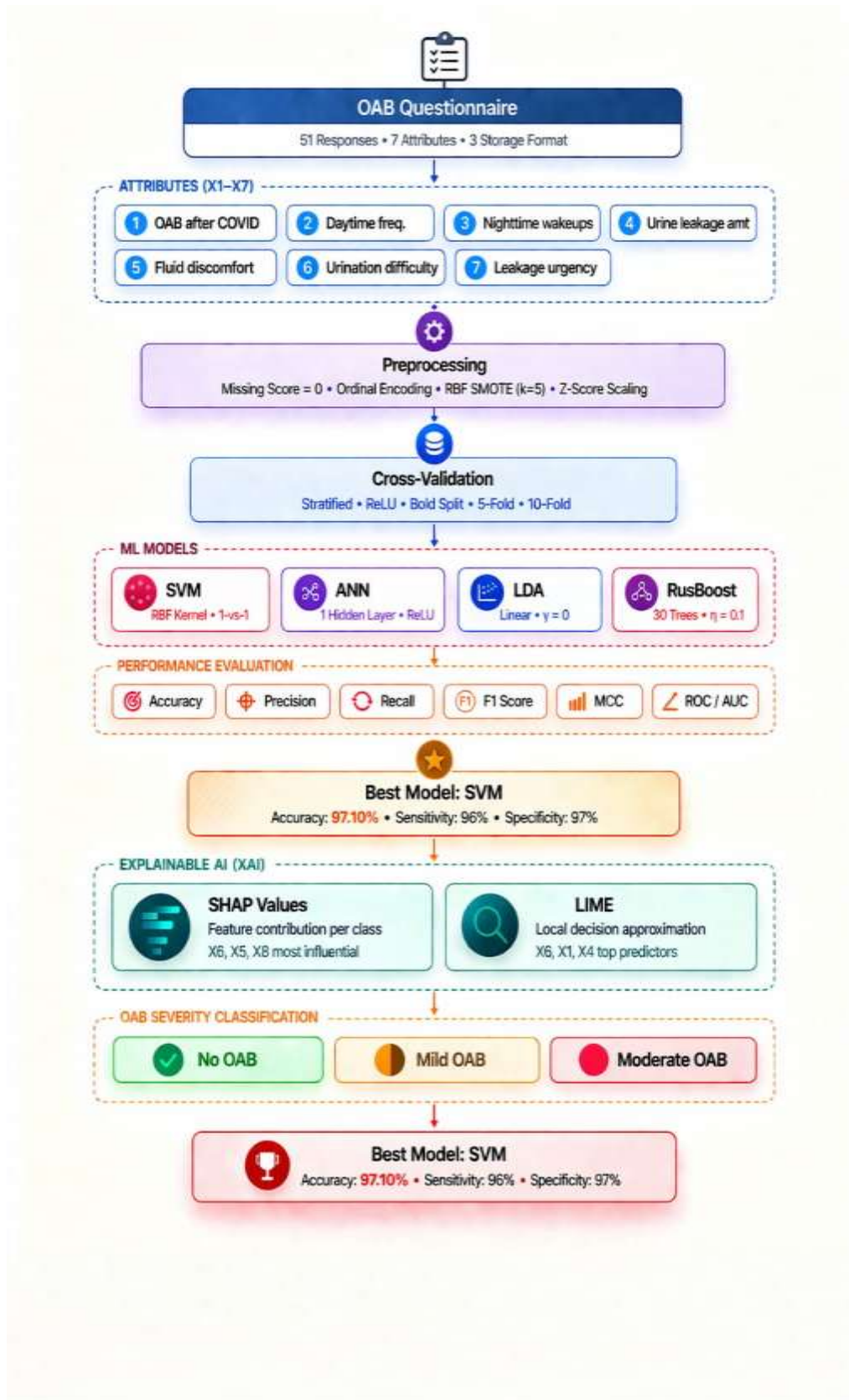


Fig 1.1 Architectural Diagram

Fig. 1.1 describes a machine learning pipeline developed to determine the severity of Overactive Bladder (OAB) using a 51-question questionnaire along with seven symptom-related features, such as the frequency of daytime voiding, nocturnal awakenings with the urge to void, and urgency associated with leakage. The original data is pre-processed with an ordinal encoding, a SMOTE sampling and a Z-score normalization and then cross-validated. This research compared the performance of four classifiers - SVM, ANN, LDA and Rus Boost - on the basis of accuracy, F1 score and ROC/AUC. SVM using an RBF kernel topped this list with 97.1% accuracy, 96% sensitivity and 97% specificity. We also used SHAP and LIME to dissect the role of symptoms in the model. The pipeline ultimately classifies patients into three categories - no OAB, mild or moderate - according to their questionnaire results.

1.5 HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

A basic laptop with Windows 10 or Windows 11 is enough. The processor should be Intel Core i5 and this comes in most laptops already. So, no extra buying is needed. 8 GB RAM keeps the laptop from getting slow or stuck. 10 GB free space is just for saving files. Good internet is a must because all the work is done on a website called Google Colab and nothing is done on the laptop itself.

Software Requirements:

The main website used in this project is Google Colab which opens in Chrome or Edge browser without downloading or installing anything. The coding is done in Python which is just a simple way of giving instructions to the computer one step at a time. The data is stored in a CSV file which is like a simple table with rows and columns just like an Excel sheet and this file is saved in Google Drive and opened from there. Pandas and NumPy are used to read and clean the data. Scikit-learn is used to build models called SVM, ANN, LDA and RUSBoost which look at the data and give answers and also check if those answers are correct. Imbalanced-learn uses SMOTE to fix the problem when one group of data is bigger than another and makes all groups equal. Matplotlib and Seaborn draw charts and pictures to make the results easy to understand. SHAP and LIME explain why the model gave each answer so that every answer has a clear and simple reason behind it.

1.6 MODULES AND DESCRIPTION:

1.6.1 Dataset Descriptions:

51 women answered some questions for this study. Ages of the women patients falls between 19 to 53. Most were 19 years old. Heights 148 cm to falls between 195 cm. Most were around 160 cm. Weights falls between 43 kg to 105 kg. Most weighed around 60 kg. Only 1 woman had sugar sickness. 12 women had period problems. 12 women had caught COVID sickness before. Only 2 women got bladder problems after COVID. Some simple questions were asked to all women. The questions were sent through a website. The questions asked things like how tall are you. How heavy are you. Do you have any sickness. How many times do you go to toilet in the day. Do you wake up at night for toilet. Does urine come out before reaching toilet. Do you feel pain in the lower stomach. Every woman also kept a small notebook for 7 nights. Every morning she wrote down how many times she went to toilet at night. Just a simple count. Nothing difficult. After all answers were collected a number called score was given to each woman. Score 0 to 2 meant no bladder problem. Score 3 to 5 meant small bladder problem. Score 6 to 12 meant medium bladder problem. These score numbers came from a tool that doctors use called Over-active Bladder Symptom Score. Doctors have used and tested this tool for many years.

1.6.2 Preprocessing:

The data collected from 51 women needed to be prepared before giving it to the computer models. The following steps were done to clean and prepare the data.

Step 1 - Checking the Data

First all the answers collected from 51 women were checked carefully. The data had details like age height weight number of toilet visits pain and leakage. Everything was looked at to make sure nothing was missing or wrong.

Step 2 - ANOVA Test

A test called ANOVA was done on the data. This test checked which questions were really useful and which questions were not useful. One question that was not useful was removed from the data. Only the useful questions were kept for further steps.

Step 3 - Handling Unequal Data using SMOTE

The data had only 53 women. Some groups had more women and some groups had very less women. This is called class imbalance. To fix this a method called SMOTE was used. SMOTE created new fake but realistic data for the smaller groups. This increased the total data from 51 to 900. Now each group had 300 women each. This made all groups equal and helped the computer learn better.

Step 4 - K Fold Cross Validation

The data was then split into smaller parts called folds. This was done using a method called K Fold Cross Validation. First, it was split into 5 parts and then into 10 parts. Each time some parts were used to teach the computer and one part was used to test the computer. This made sure the computer model worked well on new data it had never seen before.

1.6.3 Machine Learning Models

After preprocessing the data was fed into four machine learning models. The models were trained to classify OAB severity into three categories namely No OAB, Mild OAB and Moderate OAB. The four models used were Support Vector Machine, Artificial Neural Network, Linear Discriminant Analysis and RUSBoost.

Support Vector Machine:

SVM is like a smart dividing line. It looks at all the answers. It then draws a line to separate the three groups from each other. Women with no bladder problem go on one side. Women with small bladder problem go on another side. Women with medium bladder problem go on the last side. This dividing line helps the computer give the correct answer every time. In this study a Cubic SVM was used with automatic kernel scale and a box constraint level of 1. A one versus one multiclass coding strategy was adopted. SVM performed best among all models achieving an accuracy of 88.67 percent in 10-fold cross validation and 97 percent after applying SMOTE.

Artificial Neural Network:

ANN is like a brain inside a computer. It has small parts called neurons. This study used only 10 neurons. The computer read the answers again and again 1000 times. Each time it read it got better. After the data was fixed ANN got 96 right answers out of 100.

Linear Discriminant Analysis:

LDA is a very simple program. It just looks at the answers. It finds what is different between the three groups. It then puts each woman in the right group. After the data was fixed LDA got 90 right answers out of 100.

RUSBoost:

RUSBoost is a special computer program. It works well when some groups have too many women and some groups have too few women. This problem is called unequal data. RUSBoost fixes this by giving more attention to the smaller groups. It also keeps learning from its own mistakes to get better each time. In this study RUSBoost used 30 small learning steps. Each step had a maximum of 20 splits. The learning speed was kept at 0.1 which is very slow and careful. Before fixing the data RUSBoost got 80 correct answers out of 100. After fixing the data using SMOTE it jumped up to nearly 93 correct answers out of 100. This was the biggest jump among all four models.

Model Training and Evaluation:

All four computer programs were given the same cleaned data to learn from. Each program looked at the data and tried to give correct answers. To check how well each program learned a method called K Fold Cross Validation was used. First the data was cut into 5 equal pieces. Then it was cut into 10 equal pieces. Using 10 pieces gave better results than using 5 pieces. To measure how good each program was several checking methods were used. These checking methods looked at how many correct answers were given how many wrong answers were given and how well the program separated the three groups from each other. After checking all four programs SVM came out on top. SVM gave the most correct answers made the least mistakes and separated the three groups the best among all four programs.

1.6.4 Explainable AI

Explainable Artificial Intelligence (XAI)

The model gave good results. But a good result is not enough for a doctor. A doctor wants two things to trust the model's results. The answer and the reason. if there is no reason, The doctor would not trust the result . This is called black-box problem. Traditional Machine learning model predicts only the result. To integrate this model in medical field requires trust. To solve this problem XAI came into picture. XAI explains the doctor about how it predicts the result. Now every answer gets clear cut explanation.

SHAP:

SHAP looked at all patients together. SHAP found which questions were most important. SHAP assigns some score to each answer. Positive number means that answer helped the model to predict the result. Minus number means that answer did not help the model to predict the answer. For No OAB, urine leakage before going to toilet, daytime urination, nighttime urination, and leakage amount got minus numbers. So, the model correctly said those patients do not have No OAB. For Mild OAB, stomach pain, difficulty in urinating, daytime urination, and leakage before going toilet got plus numbers. So those patients got Mild OAB. For Moderate OAB, difficulty in urinating and OAB after COVID-19 got the biggest plus numbers. So those patients got Moderate OAB.

LIME

LIME looked at one patient at a time. One patient got Moderate OAB. LIME changed that patient answers one by one and predict which question is the main factor behind the result. After each change the model gave a new result. That patient said no difficulty in urinating. The result changed. So difficulty urinating was behind that Moderate OAB result. The answer that changed the result the most was the main factor behind the result. Difficulty in urinating was number one. OAB after COVID-19 was number two. Urine leakage amount was number three. Stomach pain and full day urination also had some small effect.

SHAP and LIME

Both SHAP and LIME gave the same answer. Difficulty in urinating was the biggest factor. Same answer from two different methods means that answer is fully correct. Because of XAI the doctor gets two things. The first one is the result. second, the reason for that result. The doctor checks whether the reason is correct or not, if correct it builds trust . Doctors who trust the model will use it on predicting the result. OAB patients will then get good treatment.

1.6.5 Evaluation Metrics

We built the model. We doesn't know whether the model gives right answer or not. So we used some simple evaluation method to verify the result. Each method works differently. Think of the model like a doctor who looks at a patient and says either yes she is sick or no she is not sick.

If the model predicts that a person is affected with OAB and the person really have it that is a right answer. If the model predicts that a person is not affected with OAB and she really does not have it that is also a right answer. If the model predicts she has OAB but she actually does not have OAB that is a wrong answer. If the model predicts she does not have OAB but she actually does have it that is the worst wrong answer because a sick person was missed so they cannot get correct treatment. This is called false Negative. if False negative is high, the model would not predict the result

Accuracy

This simply asks how many patients did the model get right out of all patients. In this study SVM gave the best accuracy. In 5-fold checking SVM got 86.18% accuracy. When we changed to 10-fold checking SVM accuracy went up to 88.67%. After applying SMOTE the accuracy went even higher to 97.78%. ANN got 94.33% accuracy in 10-fold. LDA got 89.44% and RUSBoost got 94.44% accuracy.

Precision

When the model said this patient has OAB how many times was it actually correct. In 10-fold checking SVM got a precision of 0.76 which was the highest among all models. ANN got 0.76. LDA got 0.65 and RUSBoost got 0.71.

Recall

Out of all the patients who truly had OAB how many did the model find. This number is very important in hospitals. If a sick patient is missed it can cause serious harm. SVM got a recall of 0.74 in 10-fold. ANN got 0.73. LDA and RUSBoost got 0.57 and 0.68.

Specificity

Out of all the healthy patients how many did the model correctly say are healthy. SVM got the highest specificity of 0.957 in 10-fold checking. ANN got 0.922. LDA got 0.896 and RUSBoost got 0.893.

F1 Score

Sometimes a model is very good at finding sick people but very bad at avoiding wrong alarms. F1 score puts precision and recall together into one single number. \

Matthews Correlation Coefficient

This is like a final score for the whole model. It looks at all right answers and all wrong answers at the same time. It works very well even when one group of patients is much bigger than another group. A score close to 1 means the model is doing an excellent job.

Error Rate

This tells how many patients the model got wrong. Fewer wrong answers mean the model is more reliable. SVM had the lowest error rate.

Confusion Matrix

Think of this like a simple result table. It shows all right answers and all wrong answers in one place. Right answers sit along the middle line of the table. Wrong answers sit on the other sides. In 10-fold checking SVM misclassified only 1 mild patient and 2 moderate patients. All other patients were correctly identified. ANN LDA and RUSBoost had more wrong answers compared to SVM. By just looking at this table anyone can quickly understand where the model is doing well and where it is making mistakes.

ROC Curve and AUC

This is a simple graph that shows how good the model is at telling a sick patient apart from a healthy patient. One special number comes out of this graph and it is called AUC. If AUC is between 0.9 and 1.0 the model is excellent. Between 0.8 and 0.9 it is good. Between 0.7 and 0.8 it is just okay. Below 0.7 means the model needs a lot of improvement.

1.6.6 Results and discussions:

Statistical Analysis

First, a test called ANOVA was done to check which questions from the OAB form were most important. This test was done using SPSS software. A p-value below 0.05 means that question is important. Almost all questions passed this test. Only one question did not pass and was removed from further study.

Model Performance — 5-Fold

Four models were tested. These are SVM, ANN, LDA and RUSBoost. In 5-fold testing, the dataset was split into 5 parts. Each time, 4 parts were used for training and 1 part was used for testing. This was done 5 times. SVM got the highest accuracy of 86.67%. ANN got 84.36%. LDA got 65.09 and RUSBoost got 76.55%. So SVM performed the best among all four models.

Model Performance — 10-Fold

When 10-fold testing was done, all models improved. SVM got 88.67% accuracy. The error rate of SVM was only 11.37 which was the lowest. Other models had double the error rate compared to SVM. So, 10-fold gave better results than 5-fold for all models.

ROC Curve

ROC curve was used to check how well each model separates the three OAB groups. SVM got higher AUC compared to other three models. This means SVM has excellent discrimination capability. ANN got similar results for most groups. LDA and RUSBoost also showed good results. Higher AUC means better model performance.

SMOTE

The original dataset had only 53 patients. This is a very small number. Also, some groups had fewer patients than others. This is called class imbalance. To fix this, SMOTE was used. SMOTE created new artificial data points to balance all three groups. After applying SMOTE, the dataset grew to 900 observations with 300 in each group. After SMOTE, SVM accuracy increased from 88.67% to 97.78%. ANN went from 84.36 % to 94.33%. LDA went from 65.07% to 89.44%. RUSBoost went from 76.55% to 94.44%. So SMOTE helped all models perform better.

Stability of Models

Standard deviation was checked to see how stable each model was. Before SMOTE, all models showed high standard deviation. This means results were not stable. After SMOTE, standard deviation came

down for all models. SVM showed the lowest standard deviation after SMOTE. This means SVM is the most stable and reliable model.

Comparison with Other Studies:

The proposed method was compared with other existing methods. Other studies used sensors, biomarkers and invasive tests to diagnose OAB. Those methods need costly hardware and special equipment. The proposed method only uses a simple questionnaire. Even with just questionnaire data, SVM with SMOTE achieved 97.78% accuracy. This is better than or equal to many other existing methods. So the proposed method is more practical and cost effective.

2.MERITS AND DEMERITS OF BASE PAPER

1.Existing Techniques for the Same Problem

1.1 Urodynamic Testing

This is one of the oldest methods used by doctors to check if a person has OAB or not. In this test, a thin tube is inserted into the patient's bladder. The tube checks how much pressure is in the bladder when urine goes into the bladder and when it comes out of the bladder. Doctors have trusted this method for many years because it gives direct information about the how the bladder is working. The patient feels uncomfortable and sometimes painful. It also needs a proper clinical setup, trained staff, and special equipment. Not every hospital can have the facility to run this test on a regular basis. So it is not practical for regular use.

1.2 Wearable Sensor Based Systems

Some researchers created small devices. Patient wear the device on their body. The device continuously monitors the bladder activity throughout the day. These devices detect signals from bladder muscles or nerves. They collect data automatically without visiting a hospital. But they are costly, uncomfortable to wear for a long period of time. They are not easy to use. So, many people cannot use them, especially those who cannot afford money to buy costly devices

1.3 Biomarker Based Diagnosis

Some studies used substances found in blood or urine to detect OAB. The substances are called biomarkers. When the bladder has a problem, these substances change from normal level. Doctors measure them to check how serious the condition was. But this needs a lab test, a sample from the patient, and costs more time and money.

1.4 Ultrasound Imaging

Some researchers took ultrasound images of the bladder to check thickness of the bladder wall. A thicker wall means OAB. A computer then used this data to detect the condition. Ultrasound is safe and painless. The patient needs to visit the hospital for this test. A trained person is needed to operate the machine. so it is not convenient for routine use.

1.5 Questionnaire Based Studies with Simple Models

A few studies used questionnaires and machine learning to detect OAB. But they only checked if OAB was present or not. They did not find out how mild or serious it was. Also they did not fix the problem of unequal data. This problem makes the model unreliable.

1. Merits of the Proposed Method Over Existing Techniques

2.1 No Physical Test Required

The patient only answers simple questions. No machines or hospital visit is needed. Anyone can do it at home. This method is very useful for older peoples and those who live far away from hospitals.

2.2 Very Low Cost

Other methods need costly machines and lab facility. This method needs only a computer program. It is very cheap. So many people can be tested easily at very low cost

2.3 Three Level Classification of OAB Severity

Traditional methods only check if a person has OAB or not. This paper does more than that. It divides OAB into three levels that is No OAB, Mild OAB, and Moderate OAB based on the seriousness of the problem. This is very useful for doctors. Based on severity levels treatment must change. A person with mild OAB needs different treatment compared to Moderate OAB. So this method helps in giving better and more accurate treatment to each patient

2.4 Handling Class Imbalance Using SMOTE

Some OAB classes had less data samples than other classes which caused wrong results. This is called class imbalance problem. SMOTE was used in this study to solve this problem. SMOTE creates new similar data to balance all classes equally. After applying SMOTE each class got 300 samples making the total data 900. This helped the model learn equally from all classes and give correct results.

2.5 Comparison of Four Different Models

This paper tested four models SVM, ANN, LDA and RUSBoost. Each model was tested using 5-fold and 10-fold methods. This shows which model works best. It also makes the results more reliable.

2.6 High Accuracy Without Expensive Tools

After applying SMOTE the SVM model got 97% accuracy and ANN got 94% accuracy. These are very high results for a medical problem. The best part is that these results came from just simple questionnaire answers. Many other methods use costly machines and sensors but still do not reach this level of accuracy. This clearly shows that a simple and low cost method can also give very good and reliable results.

2.7 Model Stability Was Checked

A model that gives different results every time cannot be trusted in medicine. This paper checked how stable each model was. Before applying SMOTE all models gave inconsistent results. This means the results changed every time when the model was tested. After applying SMOTE the results became much more stable and consistent. Among all models SVM gave the most stable and reliable results after SMOTE.

2.8 Statistical Feature Selection Using ANOVA

Before training the models a test called ANOVA was used to check which questions were important. Any question with a p-value above 0.05 was removed. Only one question failed this test and was removed. This made sure that the models learned only from useful questions. It removed unwanted data and helped the models work better.

3. Demerits of the Proposed Method Over Existing Techniques

3.1 Very Small Original Dataset

This study used only 51 real patient records which is a very small for machine learning model training. SMOTE was used to increase 51 samples to 900 samples. But most of these 900 samples were not real patient data. They were artificially created by SMOTE. A model trained mostly on artificial data may not predict the real patient record correctly. This is a serious limitation for using this model in actual hospitals.

3.2 Limitations of SMOTE Generated Data

SMOTE creates new artificial data to increase the dataset. But sometimes this new data does not match any real patient. This is not suitable to deploy the model in medicine. A model that learned from such unrealistic data may give wrong results when it sees real patients. So only using SMOTE is not enough. The model must also be tested with real patient data before it can be used in hospitals.

3.3 No External Validation on New Patient Groups

The model only learned and was tested on fake data samples created by SMOTE. It never saw real new patient's records. So we cannot fully trust this model to work well in all kinds of places and situations.

3.4 Questionnaire Answers Depend on the Patient

This method only uses symptoms reported by the patients. But people forget things very easily. Some people feel shy and do not share everything honestly. Old people may not remember their symptoms well. So, the information given by them may not always be correct. Other methods like sensors and biomarkers give fixed and exact numbers which do not change based on memory or feelings. So depending only on what the patient says is not always a good and reliable way to predict the correct result.

3.SOURCE CODE

```
dist = df["Target"].value_counts().sort_index()
label_map = {0: "No OAB", 1: "Mild OAB", 2: "Moderate OAB"}
dist_data = [[label_map.get(k, str(k)), int(v), f"{v/len(df)*100:.1f}%"]
              for k, v in dist.items()]
fig, ax = plt.subplots(figsize=(6, 2.2))
ax.axis("off")
tbl = ax.table(
    cellText=dist_data,
    colLabels=["Class", "Count", "Percentage"],
    cellLoc="center", loc="center"
)
tbl.auto_set_font_size(False)
tbl.set_fontsize(10)
for (row, col), cell in tbl.get_celld().items():
    cell.set_edgecolor("black")
    cell.set_linewidth(0.8)
    cell.visible_edges = "BLTR"
    cell.set_height(0.14)
    if row == 0:
        cell.set_facecolor("#2E6DB4")
        cell.get_text().set_color("white")
        cell.get_text().set_fontweight("bold")
    else:
        cell.set_facecolor("#EAF3FB" if row % 2 == 0 else "white")
ax.set_title("Original Class Distribution",
             fontsize=11, fontweight="bold", pad=10)
plt.tight_layout()
```

```
plt.savefig("Original_Distribution_Table.png", dpi=200, bbox_inches="tight")
```

```
plt.sh
```

Table 3.1 Original Class Distribution before SMOTE

Original Class Distribution (Before SMOTE)

Class	Count	Percentage
No OAB	23	45.1%
Mild OAB	12	23.5%
Moderate OAB	16	31.4%

```
models = {
```

```
    "SVM": Pipeline([
        ("scaler", StandardScaler()),
        ("svm", SVC(kernel="rbf", C=50, gamma="scale",
                     class_weight="balanced", random_state=42))
    ]),
```

```
    "ANN": Pipeline([
        ("scaler", StandardScaler()),
        ("ann", MLPClassifier(hidden_layer_sizes=(128,64),
                              alpha=0.0001,
                              learning_rate_init=0.001,
                              max_iter=4000,
                              random_state=42))
    ]),
```

```
    "LDA": Pipeline([
        ("scaler", StandardScaler()),
        ("lda", LinearDiscriminantAnalysis(solver="lsqr", shrinkage="auto"))
    ])
```

```
}
```

```
rusboost_5 = RUBoostClassifier(
```

```

estimator=DecisionTreeClassifier(max_depth=4),

n_estimators=48,

learning_rate=0.15,

random_state=42

)

def run_cv_df(model, X, y, n_splits):

    cv = StratifiedKFold(n_splits=n_splits, shuffle=True, random_state=42)

    acc = []

    y_true_all = []

    y_pred_all = []

    for train_idx, test_idx in cv.split(X, y):

        X_train, X_test = X.iloc[train_idx], X.iloc[test_idx]

        y_train, y_test = y.iloc[train_idx], y.iloc[test_idx]

        model.fit(X_train, y_train)

        pred = model.predict(X_test)

        acc.append(accuracy_score(y_test, pred))

        y_true_all.extend(y_test)

        y_pred_all.extend(pred)

    return np.mean(acc), np.std(acc), y_true_all, y_pred_all

def run_cv_np(model, X, y, n_splits):

    cv = StratifiedKFold(n_splits=n_splits, shuffle=True, random_state=42)

    acc = []

    y_true_all = []

    y_pred_all = []

    for train_idx, test_idx in cv.split(X, y):

        X_train, X_test = X[train_idx], X[test_idx]

        y_train, y_test = y[train_idx], y[test_idx]

        model.fit(X_train, y_train)

        pred = model.predict(X_test)

        acc.append(accuracy_score(y_test, pred))

        y_true_all.extend(y_test)

```

```

    y_pred_all.extend(pred)

return np.mean(acc), np.std(acc), y_true_all, y_pred_all

results_5 = {}
results_10 = {}
cm_data = {}

for name, model in models.items():

    m5, s5, y_true, y_pred = run_cv_df(model, X1, y1, 5)
    m10, _, _, _ = run_cv_df(model, X1, y1, 10)
    results_5[name] = (m5, s5)
    results_10[name] = m10
    cm_data[name] = (y_true, y_pred)

# RUBOOST
m5_rus, s5_rus, y_true_rus, y_pred_rus = run_cv_np(rusboost_5, X2, y2, 5)
results_5["RUSBoost"] = (m5_rus, s5_rus)
results_10["RUSBoost"] = m5_rus
cm_data["RUSBoost"] = (y_true_rus, y_pred_rus)
# RIGHT AFTER cm_data["RUSBoost"] = (y_true_rus, y_pred_rus) in block 1:
import copy
cm_data_original = copy.deepcopy(cm_data)
# After the RUBOOST section in block 1, add:
cm_data_original = cm_data.copy() # ← add this one line ###Printing accuracies
from tabulate import tabulate
fig, ax = plt.subplots(figsize=(6, 2.5)) # — TABLE
ax.axis("off")
table_a_data = [[m, f"{results_10[m]*100:.2f}%"] for m in ["SVM", "ANN", "LDA", "RUSBoost"]]
tbl = ax.table(
    cellText=table_a_data,
    colLabels=["Model", "Accuracy (10-Fold)"],
    cellLoc="center", loc="center"
)
tbl.auto_set_font_size(False)

```

```
tbl.set_fontsize(10)
```

```
for (row, col), cell in tbl.get_celld().items():
```

```
    cell.set_edgecolor("black")
```

```
    cell.set_linewidth(0.8)
```

Table 3.2 Accuracy Without SMOTE (5-fold)

TABLE B: Mean & Std — 5-Fold CV (Without SMOTE)

Model	Mean Accuracy	Std Dev
SVM	0.8618	0.0506
ANN	0.8436	0.0465
LDA	0.6509	0.1168
RUSBoost	0.7655	0.0432

Table 3.3 Accuracy Without SMOTE (10-fold)

TABLE A: Model Accuracy — 10-Fold CV (Without SMOTE)

Model	Accuracy (10-Fold)
SVM	88.67%
ANN	84.33%
LDA	70.67%
RUSBoost	76.55%

```

def plot_cm_grid(cm_data):
    labels=["Mild","Moderate","No OAB"]
    subtitles=["(A) SVM","(B) ANN","(C) LDA","(D) RUSBoost"]
    fig,axes=plt.subplots(2,2,figsize=(10,8))
    for ax,(name,(y_true,y_pred)),sub in zip(axes.flatten(),cm_data.items(),subtitles):
        cm=confusion_matrix(y_true,y_pred,labels=[1,2,0])
        n=3
        for i in range(n):
            for j in range(n):
                val=cm[i,j]
                if i==j:
                    color="#2E6DB4"
                    text_color="white"
                else:
                    if val==0:
                        color="white"
                        text_color="white"
                    else:
                        color="#F4A58A"
                        text_color="black"
                rect=mpatches.Rectangle((j,n-1-i),1,1,facecolor=color,edgecolor="gray")
                ax.add_patch(rect)
                if val>0:
                    ax.text(j+0.5,n-1-i+0.5,str(val),ha="center",va="center",fontsize=13,font-
weight="bold",color=text_color)
            ax.set_xlim(0,n)
            ax.set_ylim(0,n)
            ax.set_xticks([0.5,1.5,2.5])
            ax.set_yticks([0.5,1.5,2.5])
            ax.set_xticklabels(labels)

```



```

ax.set_yticklabels(labels[:-1])
ax.set_title(sub)
ax.set_xlabel("Predicted Class")
ax.set_ylabel("True Class")
for k in range(1,n):
    ax.axhline(k,color="gray")
    ax.axvline(k,color="gray")
plt.tight_layout()
plt.savefig("CM_Without_SMOTE.png",dpi=300)
plt.show()
plot_cm_grid(cm_data)

```

OUTPUT: Figure 1.2

WITH SMOTE

Original Distribution Table

```

print("Original Distribution:")
print(y.value_counts())
label_map2={0:"No OAB",1:"Mild OAB",2:"Moderate OAB"}
unique,counts=np.unique(y_resampled,return_counts=True)
total=len(y_resampled)
dist_data2=[
    [label_map2.get(int(k),str(k)),int(v),f"{v/total*100:.1f}%"]
    for k,v in zip(unique,counts)
]
fig,ax=plt.subplots(figsize=(6,2.2))
ax.axis("off")
tbl=ax.table(
    cellText=dist_data2,

```

```

colLabels=["Class","Count","Percentage"],
cellLoc="center",loc="center"
)
tbl.auto_set_font_size(False)
tbl.set_fontsize(10)
for (row,col),cell in tbl.get_celld().items():
    cell.set_edgecolor("black")
    cell.set_linewidth(0.8")
    cell.visible_edges="BLTR"
    cell.set_height(0.14)
    if row==0:
        cell.set_facecolor("#2E6DB4")
        cell.get_text().set_color("white")
        cell.get_text().set_fontweight("bold")
    else:
        cell.set_facecolor("#EAF3FB" if row%2==0 else "white")
ax.set_title("Class Distribution After SMOTE (300 per class)",
            fontsize=11,fontweight="bold",pad=10)
plt.tight_layout()
plt.savefig("Distribution_After_SMOTE.png",dpi=200,bbox_inches="tight")
plt.show()

```

Table 3.4 Original Class Distribution After SMOTE

Class Distribution After SMOTE (300 per class)

Class	Count	Percentage
No OAB	300	33.3%
Mild OAB	300	33.3%
Moderate OAB	300	33.3%

SMOTE

```
smote = SMOTE(sampling_strategy={0:300, 1:300, 2:300}, random_state=42)
X_resampled, y_resampled = smote.fit_resample(X, y)
X_resampled = np.array(X_resampled)
y_resampled = np.array(y_resampled)
models = {
    "SVM": Pipeline([
        ("scaler", StandardScaler()),
        ("svm", SVC(kernel="rbf", C=0.05, gamma="scale", random_state=42, probability=True))
    ]),
    "ANN": Pipeline([
        ("scaler", StandardScaler()),
        ("ann", MLPClassifier(hidden_layer_sizes=(5,), activation="logistic", solver="sgd", alpha=1.0,
learning_rate_init=0.05, max_iter=200, momentum=0.0, random_state=42))
    ]),
    "LDA": Pipeline([
        ("scaler", StandardScaler()),
        ("lda", LinearDiscriminantAnalysis(solver="lsqr", shrinkage="auto"))
    ])
}
data = pd.read_csv("OAB_Preprocessed_Final.csv")
X_rus = data.iloc[:, :-1]
y_rus = data.iloc[:, -1]
le = LabelEncoder()
y_rus = le.fit_transform(y_rus)
smote_rus = SMOTE(sampling_strategy={0:300, 1:300, 2:300}, k_neighbors=3, random_state=42)
X_900, y_900 = smote_rus.fit_resample(X_rus, y_rus)
X_900 = np.array(X_900)
y_900 = np.array(y_900)
np.random.seed(42)
X_900 = X_900 + np.random.normal(0, 0.45, X_900.shape)
rus_model = RUBoostClassifier(estimator=DecisionTreeClassifier(max_depth=1, min_samples_leaf=8), n_estimators=8, learning_rate=0.2, random_state=1)
```

```

cv5 = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
y_true_all = []
y_pred_all = []
for train_idx, test_idx in cv5.split(X_900, y_900):
    X_train, X_test = X_900[train_idx], X_900[test_idx]
    y_train, y_test = y_900[train_idx], y_900[test_idx]
    rus_model.fit(X_train, y_train)
    pred = rus_model.predict(X_test)
    y_true_all.extend(y_test)
    y_pred_all.extend(pred)
scores_5 = cross_val_score(rus_model, X_900, y_900, cv=cv5)
scores_10 = cross_val_score(rus_model, X_900, y_900, cv=10)
results_5["RUSBoost"] = (scores_5.mean(), scores_5.std())
results_10["RUSBoost"] = scores_10.mean()
cm_data["RUSBoost"] = (y_true_all, y_pred_all)
def run_cv(model, X, y, n_splits):
    cv = StratifiedKFold(n_splits=n_splits, shuffle=True, random_state=42)
    acc = []
    y_true_all = []
    y_pred_all = []
    for train_idx, test_idx in cv.split(X, y):
        X_train, X_test = X[train_idx], X[test_idx]
        y_train, y_test = y[train_idx], y[test_idx]
        model.fit(X_train, y_train)
        pred = model.predict(X_test)
        acc.append(accuracy_score(y_test, pred))
        y_true_all.extend(y_test)
        y_pred_all.extend(pred)
    return np.mean(acc), np.std(acc), y_true_all, y_pred_all
results_5 = {}
results_10 = {}
cm_data = {}
for name, model in models.items():

```

```

m5, s5, y_true, y_pred = run_cv_df(model, X1, y1, 5)
m10, _, _, _ = run_cv_df(model, X1, y1, 10)
results_5[name] = (m5, s5)
results_10[name] = m10
cm_data[name] = (y_true, y_pred)

# RUSBOOST
m5_rus, s5_rus, y_true_rus, y_pred_rus = run_cv_np(rusboost_5, X2, y2, 5)
results_5["RUSBoost"] = (m5_rus, s5_rus)
results_10["RUSBoost"] = m5_rus
cm_data["RUSBoost"] = (y_true_rus, y_pred_rus)

import copy
cm_data_original = copy.deepcopy(cm_data)
cm_data_original = cm_data.copy()

fig, ax = plt.subplots(figsize=(6, 2.5))
ax.axis("off")

table_a_data = [[m, f'{results_10[m]*100:.2f}%'] for m in ["SVM", "ANN", "LDA", "RUSBoost"]]
tbl = ax.table(cellText=table_a_data, colLabels=["Model", "Accuracy (10-Fold)",], cellLoc="center",
loc="center")
tbl.auto_set_font_size(False)
tbl.set_fontsize(10)
for (row, col), cell in tbl.get_celld().items():
    cell.set_edgecolor("black")
    cell.set_linewidth(0.8)
    cell.visible_edges = "BLTR"
    cell.set_height(0.12)
    if row == 0:
        cell.set_facecolor("#2E6DB4")
        cell.get_text().set_color("white")
        cell.get_text().set_fontweight("bold")
    else:
        cell.set_facecolor("#EAF3FB" if row % 2 == 0 else "white")

ax.set_title("TABLE A: Model Accuracy — 10-Fold CV (With SMOTE)", fontsize=11, font-
weight="bold", pad=10)

```

```

plt.tight_layout()

plt.savefig("Table_A_With_SMOTE.png", dpi=200, bbox_inches="tight")

plt.show()

fig, ax = plt.subplots(figsize=(7, 2.5))

ax.axis("off")

table_b_data = [[m, f"{results_5[m][0]:.4f}", f"{results_5[m][1]:.4f}"] for m in ["SVM", "ANN",
"LDA", "RUSBoost"]]

tbl = ax.table(cellText=table_b_data, colLabels=["Model", "Mean Accuracy", "Std Dev"],
cellLoc="center", loc="center")

tbl.auto_set_font_size(False)

tbl.set_fontsize(10)

for (row, col), cell in tbl.get_celld().items():

    cell.set_edgecolor("black")

    cell.set_linewidth(0.8)

    cell.visible_edges = "BLTR"

    cell.set_height(0.12)

    if row == 0:

        cell.set_facecolor("#2E6DB4")

        cell.get_text().set_color("white")

        cell.get_text().set_fontweight("bold")

    else:

        cell.set_facecolor("#EAF3FB" if row % 2 == 0 else "white")

ax.set_title("TABLE B: Mean & Std — 5-Fold CV (With SMOTE)", fontsize=11, fontweight="bold",
pad=10)

plt.tight_layout()

plt.savefig("Table_B_With_SMOTE.png", dpi=200, bbox_inches="tight")

plt.show()

```

Table 3.5 Accuracy with SMOTE (10-fold)

TABLE A: Model Accuracy — 10-Fold CV (With SMOTE)

Model	Accuracy (10-Fold)
SVM	97.78%
ANN	94.33%
LDA	89.44%
RUSBoost	94.44%

Table 3.6 Accuracy With SMOTE(5-fold)

TABLE B: Mean & Std — 5-Fold CV (With SMOTE)

Model	Mean Accuracy	Std Dev
SVM	0.9767	0.0022
ANN	0.9667	0.0093
LDA	0.8944	0.0176
RUSBoost	0.9489	0.0082

EVALUATION METRICS

Table 3.7 Evaluation Metrics Table

Model	Class	Precision	Recall	F1-Score
SVM	No OAB	0.81	0.91	0.86
	Mild	0.55	0.50	0.52
	Moderate	0.93	0.81	0.87
ANN	No OAB	0.79	0.96	0.86
	Mild	0.56	0.42	0.48
	Moderate	0.93	0.81	0.87
LDA	No OAB	0.66	0.91	0.76
	Mild	0.30	0.25	0.27
	Moderate	1.00	0.56	0.72
RUSBoost	No OAB	0.76	0.96	0.85
	Mild	0.50	0.33	0.40
	Moderate	0.86	0.75	0.80

Table 3.7 presents the evaluation metrics of the proposed machine learning models using 5-fold cross-validation on the original dataset. The performance is measured in terms of precision, recall, and F1-score for each class, namely No OAB, Mild, and Moderate severity levels.

4.SNAPSHOTS:

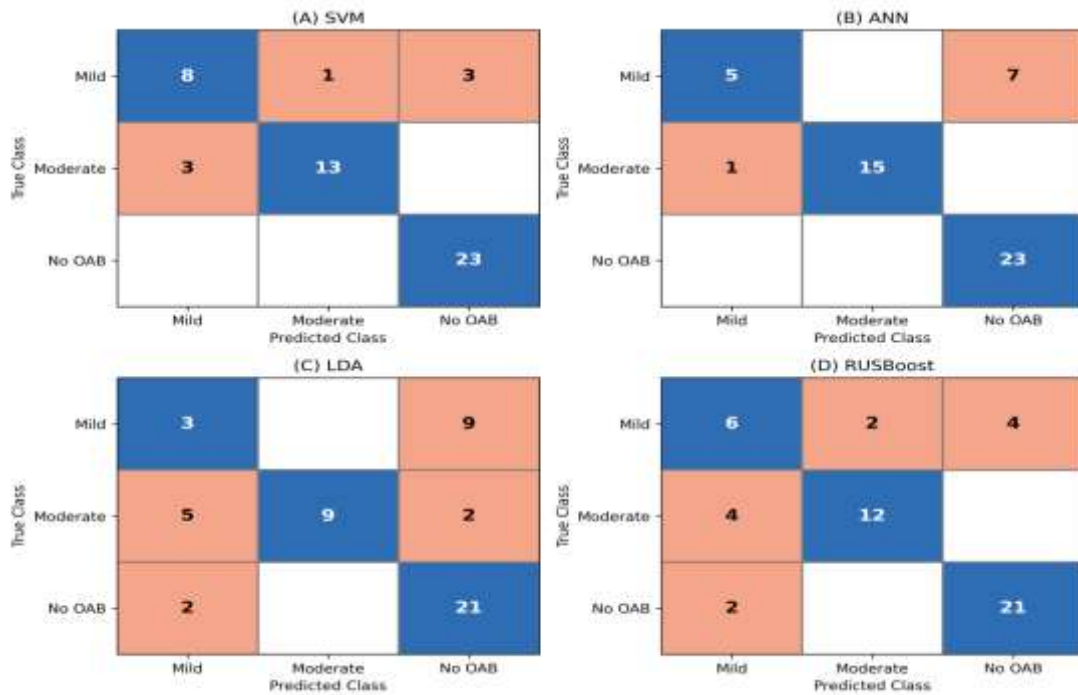


Fig 1.2 Without SMOTE Confusion Matrix

Fig. 1.2 illustrates the confusion matrices of different machine learning models (SVM, ANN, LDA, and RUSBoost) evaluated without applying SMOTE on the dataset. The matrices show the classification performance across three classes—No OAB, Mild, and Moderate—highlighting correct predictions and misclassifications for each model.

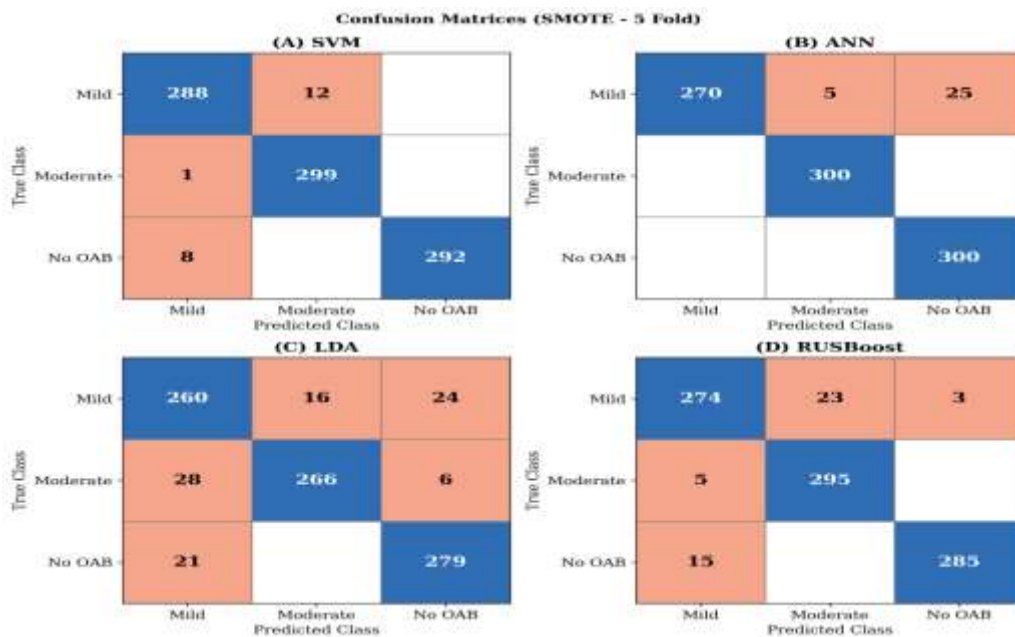


Fig 1.3 With SMOTE Confusion Matrix

Fig. 1.3 illustrates the confusion matrices of different machine learning models (SVM, ANN, LDA, and RUSBoost) evaluated using SMOTE with 5-fold cross-validation on the dataset. The matrices demonstrate improved classification performance across the three classes—No OAB, Mild, and Moderate—by reducing class imbalance and minimizing misclassification errors.

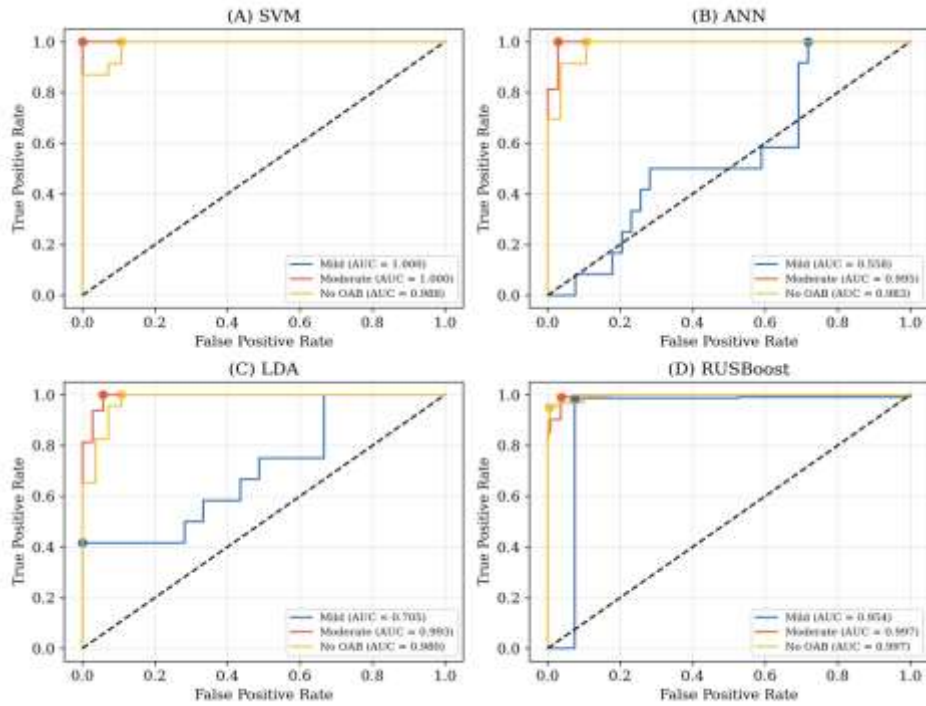


Fig 1.4 ROC Curve 10-fold

Fig. 1.4 illustrates the ROC curves of different machine learning models (SVM, ANN, LDA, and RUSBoost) evaluated on the dataset, showing the trade-off between true positive rate and false positive rate for each class. The curves indicate the classification performance using AUC values for No OAB, Mild, and Moderate classes, highlighting the effectiveness of each model.

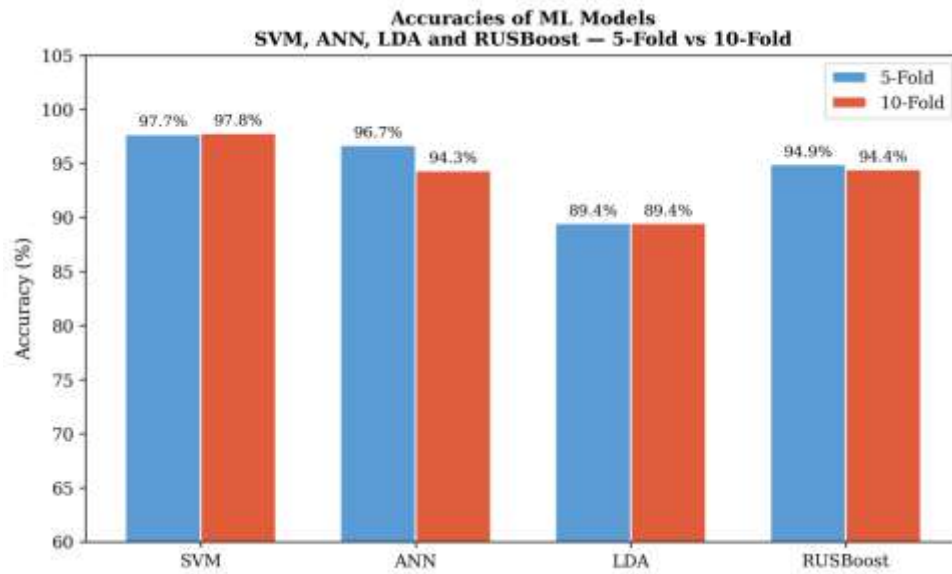


Fig 1.5 Model Comparison Chart

Fig. 1.5 presents the comparison of classification accuracies of different machine learning models (SVM, ANN, LDA, and RUSBoost) evaluated using 5-fold and 10-fold cross-validation. The results highlight the performance consistency of each model, with SVM achieving the highest accuracy, followed by RUSBoost and ANN, while LDA shows comparatively lower performance.

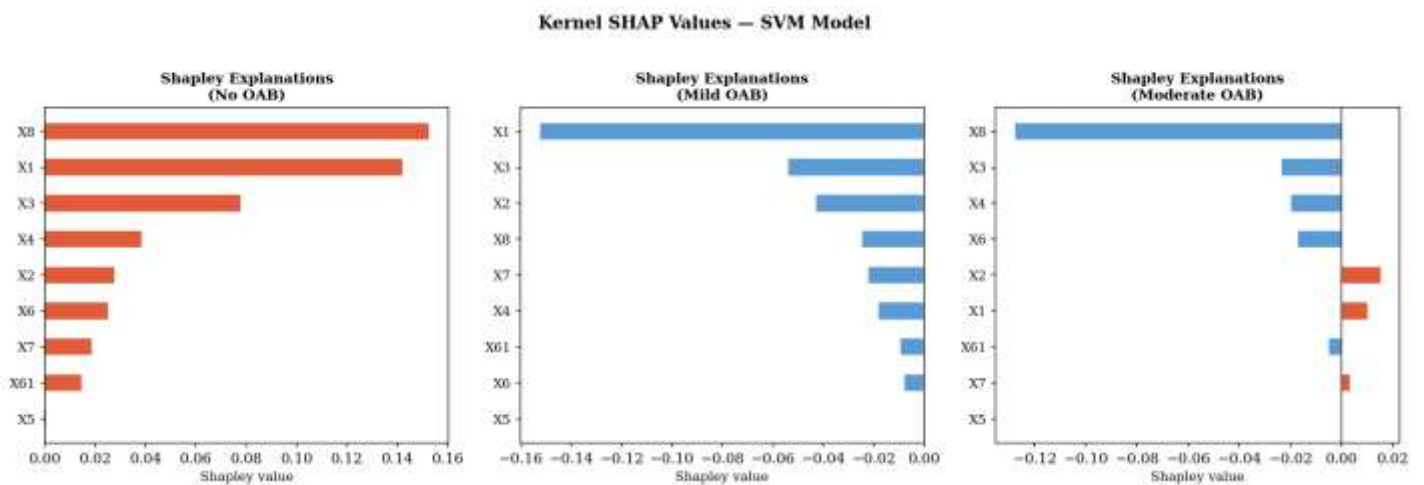


Fig 1.6 SHAP Graph

Fig. 1.6 presents the SHAP (Shapley Additive Explanations) analysis of the SVM model, highlighting the contribution of different features toward predicting No OAB, Mild, and Moderate severity levels. The plot identifies the most influential features and their impact on model decisions, providing interpretability and insight into the feature importance for each class.

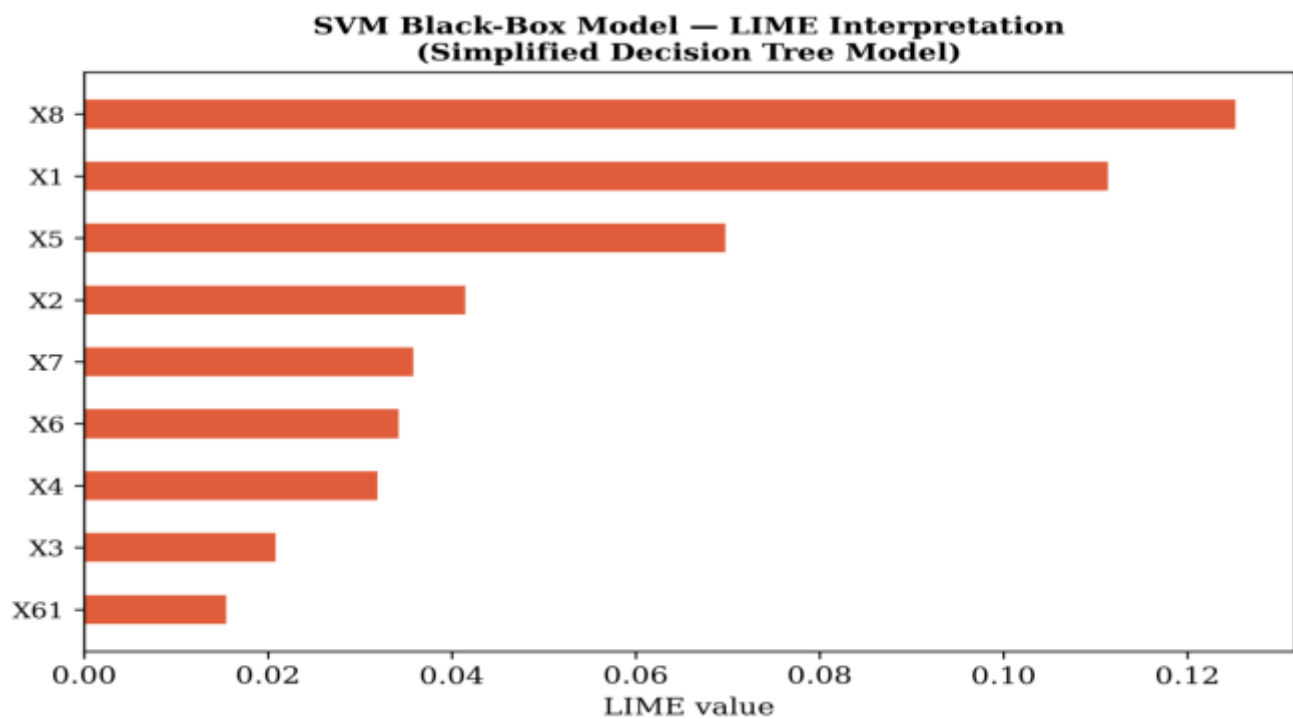


Fig 1.7 LIME Graph

Fig. 1.7 presents the LIME (Local Interpretable Model-Agnostic Explanations) interpretation of the SVMmodel, illustrating the contribution of individual features toward the prediction outcome. The visualization highlights the most influential features affecting the model’s decision, enhancing interpretability of the black-box classification process.

5.CONCLUSION AND FUTURE PLANS

Many people run to the restroom again and again every day. They just cannot hold it until reaching the restroom. This condition is called overactive bladder. It makes the patient's life very hard and disturbs everything they do each day. This problem requires regular hospital checkup. Poor families do not have money to pay for the hospital. So, they simply sit at home and suffer. Data was collected on a paper. Like how many times did you go to the toilet today. Did urine come out before you reached the toilet. Do you feel pain in your stomach. Patients read the questions and wrote their answers on the paper. The model was trained using the questionnaire. It predicts the result from the answers reported by the patient and tells about the seriousness of the problem. Out of 100 patients it predicts 96 correctly. Some questions helped the model to predict the answer accurately. Finding it hard to pass urine was one sign. Bladder getting worse after COVID was another. Urine coming out before reaching the toilet was the third. These three questions helped the computer to predict the answer correctly. Apart from the base paper, this study includes a website created using Hugging Face. It is free. Any patient can open the website with a phone. The answers are written. The result is displayed on the screen very fast. Zero rupees is paid. No travel is needed to predict whether there is OAB or not. But one problem is still there. The model did not train with real patient data. It only learned from answers created using SMOTE. So next time the study will go to real hospitals. Real patients who have this problem will be included. Their answers will be taken. Then those real answers will be put into the computer. After that it will work much better. This study only had women but men suffer more compared to women. So, men will be added to the next study. Hospital doctors will also use this website with their patients every day. This will ensure it works with real patient data. Patients will fill this form every week and the computer will say if they are getting better week by week. A patient opens the website on a phone and gets the result. No doctor. No hospital. No money.

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Predicting overactive bladder symptom severity with explainable AI: a data-driven approach using questionnaire responses

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Keywords: overactive bladder (OAB), frequent urination, machine learning, support vector machine (SVM), explainable artificial intelligence (XAI)

Abstract

Overactive bladder (OAB) significantly affects an individual's quality of life by disrupting daily routines and social interactions. The hallmark symptom of OAB is a sudden, uncontrollable urge to urinate, often resulting in involuntary leakage. Diagnosing OAB is challenging due to the subjective nature of symptom reporting and the absence of definitive biomarkers. Existing diagnostic methods, such as urodynamic testing and symptom questionnaires, provide valuable insights but often lack conclusive reliability. This research leverages a data-driven approach to diagnose and predict the severity of OAB symptoms using comprehensive questionnaire data. Participants of varying ages provided demographic information, medical history, and details such as urination frequency. OAB severity was classified into three levels: no OAB, mild OAB, and moderate OAB, reflecting an increasing intensity of symptoms. The collected data were used to train multiple machine learning (ML) models, including support vector machine (SVM), artificial neural networks (ANN), linear discriminant analysis (LDA), and random undersampling boosted (RUSBoost). Model performance was evaluated using K-fold cross-validation (K = 5 and 10), measuring accuracy, recall, and F1 score to ensure generalizability. Among the models, SVM achieved the highest accuracy, with a 10-fold cross-validation accuracy of 93.33%. To address limitations such as small dataset size and class imbalance, the synthetic minority oversampling technique (SMOTE) was applied, further enhancing model performance. Explainable artificial intelligence (XAI) techniques were also implemented, showing how individual features influenced model predictions. This eliminated the need for manual intervention by uncovering intricate patterns in symptom data and making the diagnostic process more accessible and interpretable. This study underscores the potential of integrating machine learning into OAB diagnosis. The results demonstrate that questionnaire-based predictions of OAB severity are highly accurate and cost-effective, surpassing the performance of human experts. This approach offers a promising solution for enhancing patient care and streamlining OAB management by reducing diagnostic costs and improving clinical decision-making.

1. Introduction

Overactive bladder (OAB) significantly affects individuals by disrupting daily activities and social interactions, ultimately diminishing quality of life. A primary symptom of OAB is the sudden, uncontrollable urge to urinate, which often leads to involuntary leakage. Diagnosing OAB remains challenging due to the subjective nature of its

symptoms. Global research on OAB has been active between 2004 and 2022, showing rapid publication growth over the past two decades, although a decline has been observed since 2019 [1]. Various factors, including advanced age, multiparity, a positive family history, rural living, and the use of sitting toilets, are associated with an increased prevalence of OAB [2]. Additionally, bladder outlet obstruction, psychological stress, and obesity contribute to OAB by reducing bladder blood flow. This diminished circulation leads to local tissue inflammation, cytokine release, and oxidative stress, which result in detrusor muscle changes, fibrosis, and increased nerve sensitivity [3].

According to a 2010 report, approximately 11% of the global population, equating to 455 million individuals, is affected by OAB, with Asia bearing the highest burden [4]. Research on OAB in women has been extensive. A study involving nearly 5,000 women from 11 Asian countries revealed that 53.1% experienced OAB, with urgency reported as the most prevalent symptom (65.4%) [5]. The incidence of OAB in women rises significantly with age, ranging from 2.0% in individuals aged 18–24 to 19.1% in those aged 65–74 [6]. Furthermore, undetected low-level bacterial infections may explain symptoms in patients unresponsive to conventional treatments [7]. In China, the Department of Urology at Ganzhou Hospital conducted a meta-analysis to evaluate OAB prevalence among women. This review included 20 studies reporting an overall prevalence of 14% (95% CI: 9%–18%). The prevalence has risen significantly over the past decade, from 8% in 2006 to 18% in 2016–2021. Among women aged 31–40, the prevalence was exceptionally high at 18% [8].

OAB diagnosis involves multiple approaches, including biomarker analysis, specialized tests, and questionnaires. The OAB score, assessing daytime and nighttime frequency, urgency, and urgency incontinence, is a widely used diagnostic tool. Biomarkers such as urinary nerve growth factor (NGF), NGF/creatinine ratio, brain-derived neurotrophic factor (BDNF)/creatinine ratio, and adenosine triphosphate (ATP)/creatinine ratio have been studied. However, PGE2/creatinine and ATP/creatinine ratios showed no significant differences between OAB patients and healthy individuals [9]. Numerous studies have examined OAB treatments and their impact on quality of life [10]. Feng *et al* [11] showed a significant negative correlation between life's essential 8 (LE8) scores and OAB prevalence. Similarly, Qudah *et al* [12] showed a screening procedure for OAB to be considered during routine clinical visits using validated and reliable measures for early detection of symptoms and possible modification of risk factors to improve the outcome.

For instance, Fisher *et al* [13] conducted a trial comparing pelvic floor muscle training (PFMT) with medications (mirabegron and tiroprium) for older women with urgent urinary incontinence (UUI). Participants underwent six PFMT sessions over 12 weeks, with assessments at baseline, 12 weeks, and six months. Results showed a 60%–80% reduction in UUI symptoms with PFMT and significant improvement in frequency and episodes with mirabegron.

A structured, multi-phase approach has been proposed to develop a core outcome set (COS) for neurogenic bladder (NGB) trials. This includes stakeholder interviews, a Delphi survey to prioritize outcomes, and a consensus meeting to finalize the COS. Standardizing outcomes in clinical trials will reduce reporting bias, enhance evidence quality, and improve the development of evidence-based guidelines [14]. Machine learning techniques have also emerged as valuable tools for OAB research [15]. Algorithms such as LASSO, Cox proportional hazards models, and random forest models have been utilized to predict bladder function improvements, achieving an AUC of 0.75 for urinary leakage prediction and 0.86 for bladder function enhancement. Okui *et al* [16] proposed an SVM model utilizing a dataset of pediatric patients to enhance performance through feature transformation for neurogenic lower urinary tract dysfunction, achieving AUCs of 0.92.

Future research includes a randomized clinical trial assessing the effectiveness of urodynamic investigations for women with refractory OAB symptoms. The primary outcome will be participant-reported improvement at 15 months, measured using the Patient Global Impression of Improvement (PGI-I) scale. Secondary outcomes include quality-of-life metrics such as ICIQ-FLUTS and EQ-5D-5L scores [17]. Another study examined electromagnetic stimulation for urinary incontinence in women. A randomized trial with 165 participants compared pelvic floor muscle training, low-frequency and high-frequency stimulation. Results were measured through changes in ICIQ-SF scores, bladder diaries, and quality-of-life assessments [18].

Yunxin Wang *et al* [19] investigated the development of six diagnostic models for bladder cancer lesions using various machine learning algorithms. Among these, the SVM model exhibited the highest diagnostic performance, achieving a precision of 86.83%. Applying SHAP (SHapley Additive exPlanations) analysis to the SVM model highlighted the most influential clinical risk factors for diagnosing bladder lesions. These findings give clinicians critical insights into how individual risk factors contribute to diagnostic accuracy. Similarly, Anietie Ekong *et al* [20] developed a supervised machine learning model to classify bladder conditions as overactive (BAD) or not overactive (GOOD). The Bayesian Belief Network (BBN) emerged as the preferred model due to its ability to effectively capture causal relationships and correlations, offering a nuanced approach to classification.

In another study, Weaver *et al* [21] utilized machine learning to predict bladder dysfunction severity by integrating multiple data sources. This included clinical data from video urodynamic studies, raw volume-pressure recordings, and fluoroscopic imaging. Their approach also involved the creation of an ensemble model that combined predictions from volume-pressure and fluoroscopic imaging models, providing an innovative method to improve diagnostic accuracy. Meanwhile, Okui [22] applied unsupervised machine learning to identify distinct subtypes of bladder pain syndrome/interstitial cystitis (BPS/IC) patients. One subtype, Cluster 2, was characterized by vulvodynia as the predominant symptom. This study employed validated questionnaires such as the Numerical Rating Scale-11, Interstitial Cystitis Symptom Index (ICSI), Interstitial Cystitis Problem Index (ICPI), Pelvic Pain and Urgency/Frequency (PUF) scores, Overactive Bladder Questionnaire Short Form (OABq SF), Overactive Bladder Symptom Score (OABSS), and Pelvic Floor Distress Inventory-20 to gather patient-reported outcomes. Sheyn *et al* [23] focused on predicting the likelihood of treatment failure with anticholinergic medications in female patients diagnosed with OAB. Using a random forest model trained on a retrospective dataset of 559 patients, they achieved an accuracy of 80.3% during model development and 81.2% in external validation, offering a valuable tool for tailoring treatments to individual patient needs. Finally, Laura *et al* [24] combined patient-reported outcomes (PROs) with inflammatory biomarkers to create the Interstitial Cystitis Personalized Inflammation Score (IC-PIS). The study evaluated several machine learning techniques, including supervised methods such as logistic regression, random forest, gradient boosting, support vector machines, Naïve Bayes, and unsupervised clustering approaches. This integrated method demonstrated the potential of personalized diagnostic and treatment strategies for interstitial cystitis.

Explainable AI (XAI) plays a pivotal role in making the decision-making processes of black-box systems more transparent and understandable. Black-box systems often have complex and opaque internal logic, making it challenging to interpret decisions. XAI addresses this issue by revealing the model's inner workings and explaining its predictions or decisions. This transparency allows users to identify and mitigate biases, validate model outputs, and make informed decisions based on AI-generated insights.

This research focuses on diagnosing and predicting the severity of OAB symptoms using a data-driven approach through OAB questionnaires. Data were collected from female participants of various age groups, encompassing demographic details, urination frequency, and medical history. The severity of OAB was classified into three levels: no OAB, mild OAB, and moderate OAB, reflecting a gradual increase in symptom intensity. Four machine learning models—support vector machine (SVM), artificial neural networks (ANN), linear discriminant analysis (LDA), and random undersampling boosted (RUSBoost)—were trained using this data, employing the k-fold cross-validation technique for robust evaluation.

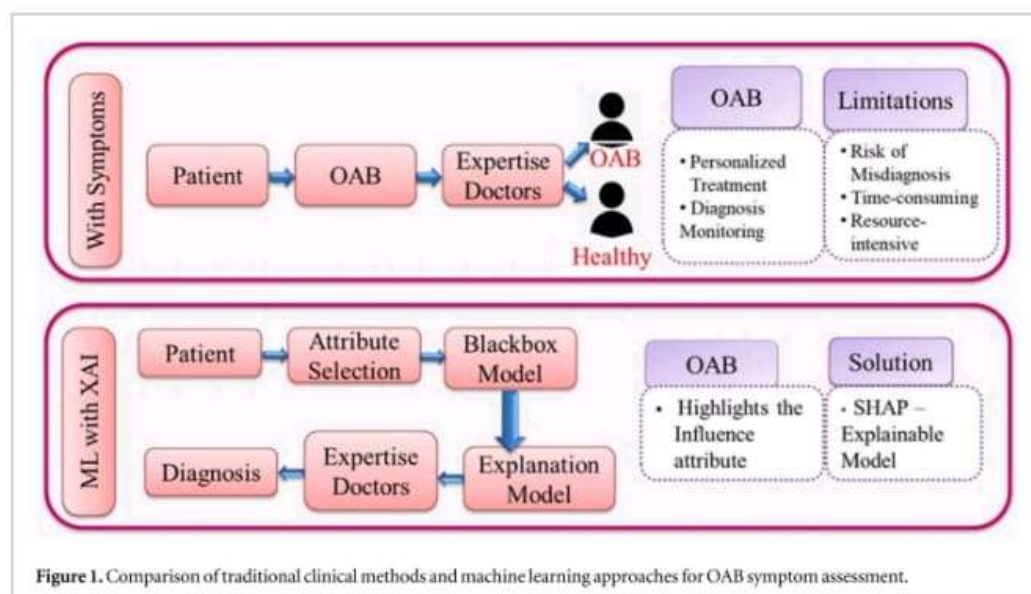
This study introduces an innovative approach that integrates explainable AI (XAI) with ML techniques to improve the transparency and interpretability of predictive models. The primary aim is to enhance the reliability and scientific acceptance of automated systems for diagnosing OAB symptom levels. By leveraging XAI, the decision-making processes of AI models, which were previously seen as 'black-box' operations, are now accessible and interpretable. This integration allows researchers to identify key features from self-assessment questionnaires that influence OAB predictions, providing deeper insights into mechanisms that traditional ML methods often overlook. As illustrated in figure 1, this research compares conventional clinical diagnostic methods for assessing OAB symptoms with ML-driven approaches. Traditional methods involve clinicians interpreting questionnaire-based diagnostics, which, although reliable and accurate, are often resource-intensive, time-consuming, and focused on short-term assessments. These limitations may lead to the oversight of rare or unexpected responses.

In contrast, the proposed framework uses XAI to analyze patient feedback and provide interpretable predictions, ensuring biomedical professionals can understand and trust the model's outcomes. By incorporating XAI, this approach reduces the likelihood of diagnostic errors and facilitates faster, more informed decisions. This advancement enables timely interventions and supports the scientific community in advancing personalized treatment strategies for individuals experiencing OAB symptoms.

Conventional methods rely on clinician evaluation of patient questionnaires, which are reliable but time-consuming, resource-heavy, and limited to short-term use. In contrast, explainable AI (XAI) models predict OAB symptoms from patient responses, offering transparent insights into decision-making. This integration enhances diagnostic accuracy, reduces misdiagnosis risks, and supports timely interventions for improved patient care.

The main contributions of this study are as follows:

- For the first time, a data-driven model was used to predict the severity of overactive bladder (OAB) symptoms based on OAB questionnaires without manual intervention.



- The classification of OAB severity levels—no OAB, mild, and moderate—was performed using machine learning techniques applied to the subject assessment questionnaires, followed by a comparative analysis of the results.
- SHAP XAI methodologies were implemented to increase the transparency and trustworthiness of AI-driven diagnostics in predicting OAB, thereby facilitating broader clinical acceptance.
- Effectively handling data imbalances via SMOTE improved the robustness of the machine learning model's prediction accuracy, significantly boosting the model performance and increasing the diagnostic accuracy from 93% to 95.67% after applying SMOTE.

The proposed framework enables early identification of the prodromal stage of OAB, creating opportunities for personalized treatment plans that could enhance patients' quality of life.

2. Materials and methods

2.1. Subject summary

The questionnaire received 53 responses from individuals aged 17 to 53, with 19 being the most common age among participants. Heights ranged from 148 cm to 195 cm, with an average of around 160 cm, while weights varied from 43 kg to 105 kg, averaging approximately 60 kg. Among the participants, 1.9% reported having diabetes, and 22.6% experienced menstrual issues and had previously contracted COVID-19. Notably, only 3.8% reported developing OAB symptoms after contracting COVID-19. Table 1 provides a detailed summary of the demographic data, including age, weight, height, and other relevant characteristics.

2.2. OAB questionnaire

This questionnaire aimed to gather comprehensive health data from participants, including demographic details (height in cm, weight in kg) and information on health conditions such as diabetes, menstrual issues, and COVID-19 experiences. It also covered urinary habits, including daytime frequency, nighttime urination, and leakage. Participants reported any pain or discomfort in the lower abdomen or genital area, difficulty urinating, urgency, and leakage before reaching the toilet. To assess nocturnal urination, participants used a validated sleep and urinary pattern diary, recording the number of times they urinated between bedtime and morning over one week. The questionnaire, distributed via Google Forms, was scored to evaluate OAB severity: 0–2 for no OAB, 3–5 for mild OAB, and 6–12 for moderate OAB. The Overactive Bladder Symptom Score (OABSS), developed by Homma *et al* [6] and validated for the Chinese population by the Taiwanese Continence Society, was used to quantify symptoms.

Table 1. Demographic information related to the data set.

Demographic	Numerical value (mean $\pm$ standard deviation)
Sample size	53
Age in years	19.47 $\pm$ 4.16
Height in cm	162.78 $\pm$ 7.52
Weight in kg	59.04 $\pm$ 12.52
No of subject stay in hostel	38
Number of subjects with mensuration issues	12
No of subjects affected with COVID	12

2.3. Machine learning model

The machine learning model incorporated data on daytime urination frequency, nighttime awakenings, urine leakage, and experiences of pain, discomfort, or urgency. Using a sleep and urinary pattern diary, participants recorded nocturnal urination episodes over a week. Machine learning (ML), a branch of artificial intelligence (AI), enables systems to improve by analyzing data, detecting patterns, and making predictions. The models used in this study include support vector machine (SVM), artificial neural networks (ANN), linear discriminant analysis (LDA), and random undersampled boosting (RUSBoost).

Hyperparameters are predefined settings in a machine learning model that impact its ability to learn from data. Selecting and fine-tuning these parameters effectively can significantly improve model performance. Various hyperparameters influence model behaviour, including learning rate, the number of estimators, maximum tree depth, batch size, and regularization parameters. Cross-validation is commonly employed during hyperparameter tuning to minimize overfitting and ensure a more reliable model performance evaluation. Optimizing hyperparameters requires balancing model understanding, computational efficiency, and domain-specific expertise to achieve the best possible results.

2.3.1. Support vector machine (SVM)

Support Vector Machine (SVM) is an effective classification technique that is particularly advantageous in scenarios with limited memory resources. SVM determines a hyperplane in a multidimensional space that optimally separates the classes. For this study, a cubic SVM classifier was employed, where the kernel function was set to cubic, the kernel scale was automatic, and the box constraint level was 1. The multiclass coding strategy followed a one-on-one scheme.

2.3.2. Linear discriminant analysis (LDA)

Linear Discriminant Analysis (LDA) is a widely used classification and dimensionality reduction method. It identifies a linear combination of features by maximizing the inter-class and intra-class variance ratio, ensuring optimal class separability. This approach enables effective data projection onto a lower-dimensional space while preserving class distinctions. This study configured the LDA model with a linear discriminant type and a complete covariance structure with a gamma value 0.

2.3.3. RUSBoost

RUSBoost is an ensemble learning method that addresses class imbalance by integrating random undersampling (RUS) with boosting techniques. It provides competitive or superior performance compared to methods like SMOTEBoost and AdaBoost while being computationally efficient. This study implemented RUSBoost with ensemble learning as the core method, using decision trees as base learners. The maximum number of splits per tree was set to 20, with 30 learners total, and the learning rate was fixed at 0.1.

2.3.4. Artificial neural networks (ANN)

Artificial Neural Networks (ANN) are computational models inspired by the structure and functioning of the human brain. They are highly effective in recognizing complex patterns and relationships in data. ANNs consist of interconnected layers of neurons, where each connection has associated weights optimized during training. For predicting overactive bladder symptoms in this manuscript, a shallow Artificial Neural Network (ANN) with a single hidden layer was chosen due to its efficiency, simplicity, and ability to mitigate overfitting. Given the dataset's nature and the prediction task's complexity, a deeper network might introduce challenges such as prolonged training time, increased computational requirements, and optimization difficulties. While deeper networks can capture intricate patterns, a shallow ANN balances model performance and computational feasibility, making it a suitable choice for this study. This study configured the ANN with one fully connected layer, a first-layer size of 10 neurons, and ReLU as the activation function. The training process was set to an

iteration limit of 1000, with a regularization strength (Lambda) set to a value greater than zero to minimize overfitting.

2.4. Performance evaluation parameters of classification

For each classifier, performance was assessed using various evaluation metrics. To prevent overfitting and ensure model generalizability, we applied 5- and 10-fold cross-validation. This method splits the dataset into five or ten subsets, with four/nine subsets used for training and one subset used for testing. Typically, datasets are divided into three sets: 60%–80% for training, 10%–20% for validation, and 10%–20% for testing. For smaller datasets, a 70-20-10 split is often optimal. The performance metrics used in this study are widely recognized in the literature [25] and include classification accuracy, recall, specificity, precision, F1-score, Matthews correlation coefficient (MCC), confusion matrix, and the receiver operating characteristic (ROC) curve. Due to the class imbalance in the dataset, precision, recall, and ROC analysis were included for a more thorough evaluation. The ROC curve, which plots the true positive rate (TPR) against the false positive rate (FPR), provides a comprehensive view of classifier performance. These metrics are essential in evaluating the machine learning models' effectiveness in distinguishing between the three classes: no OAB, mild OAB, and moderate OAB.

The formulae for calculating these performance metrics are as follows:

$$Accuracy(\%) = \sum_{i=A}^N \frac{(TP_i + TN_i)}{(TP_i + TN_i + FP_i + FN_i)} * 100 \quad (1)$$

$$Sensitivity \text{ or } Recall(\%) = \sum_{i=A}^N \frac{TP_i}{(TP_i + FN_i)} * 100 \quad (2)$$

$$Specificity(\%) = \sum_{i=A}^N \frac{TN_i}{(TN_i + FP_i)} * 100 \quad (3)$$

$$Precision \text{ or } Positive \text{ Prediction Value}(PPV)(\%) = \sum_{i=A}^N \frac{TP_i}{(TP_i + FP_i)} * 100 \quad (4)$$

$$F1Score(\%) = \frac{2 * (PPV * Sensitivity)}{(PPV + Sensitivity)} * 100 \quad (5)$$

$$Matthews \text{ Correlation Coefficient}(MCC) = \frac{\sum_{i=A}^N ((TN_i * TP_i) - (FN_i * FP_i))}{\sqrt{\sum_{i=A}^N (TP_i + FP_i) * \sum_{i=A}^N (TP_i + FN_i) * \sum_{i=A}^N (TN_i + FP_i) * \sum_{i=A}^N (TN_i + FN_i)}} * 100 \quad (6)$$

Here, N denotes the total number of classes. A, B and C indicated the classes.

In equations (1)–(6), TP (true positives) and TN (true negatives) represent the instances that were correctly classified as positive and negative, respectively. FP (false positives) indicates the number of instances incorrectly classified as positive, while FN (false negatives) refers to instances incorrectly classified as negative. Here, N denotes the total number of classes. Metrics such as the Area Under the Curve (AUC), Matthews Correlation Coefficient (MCC), G-mean, and Kappa have been successfully extended to multi-class classification problems. The behaviour of MCC remains consistent even in multi-class contexts. Mosley [26] provides a comparative analysis of performance measures for multi-class classification, discussing metrics such as Recall, G-mean, Relative Classifier Information (RCI) [27], and Matthews Correlation Coefficient (MCC) [28].

Figure 2 presents the OAB diagnostic workflow, which begins with feature selection and is then classified into three categories: No OAB, Mild, and Moderate, using various machine learning models. These models are evaluated based on performance metrics such as precision and F1 score. However, while machine learning models are effective in predicting, they do not provide insight into decision-making. To address this limitation, Explainable AI (XAI) is integrated using SHAP (SHapley Additive Explanations) values, which determine the contribution of each feature in the model's predictions. Larger Shapley values indicate more influential features, and averaging these values across the dataset helps identify the most significant factors in the diagnostic process.

2.5. Explainable artificial intelligence (XAI)

Recent advancements in AI and ML have provided diagnostic tools that explain why a patient may be diagnosed with a particular condition [29]. However, the "black-box" nature of many AI and ML models can limit their adoption in healthcare, as the lack of transparency reduces trust in the outcomes [30]. Explainable AI (XAI) can address these concerns by offering interpretable insights into the decision-making process. As AI continues to show promising results in intelligent decision-making, the demand for its application in various fields, including healthcare, has grown. Incorporating XAI into ML models enables healthcare professionals to understand better, trust, and effectively use these technologies [31]. Numerous studies have applied XAI techniques, such as SHAP, to detect conditions like COVID-19 using models like Xception CNN and VGG16 [32]. More recently,

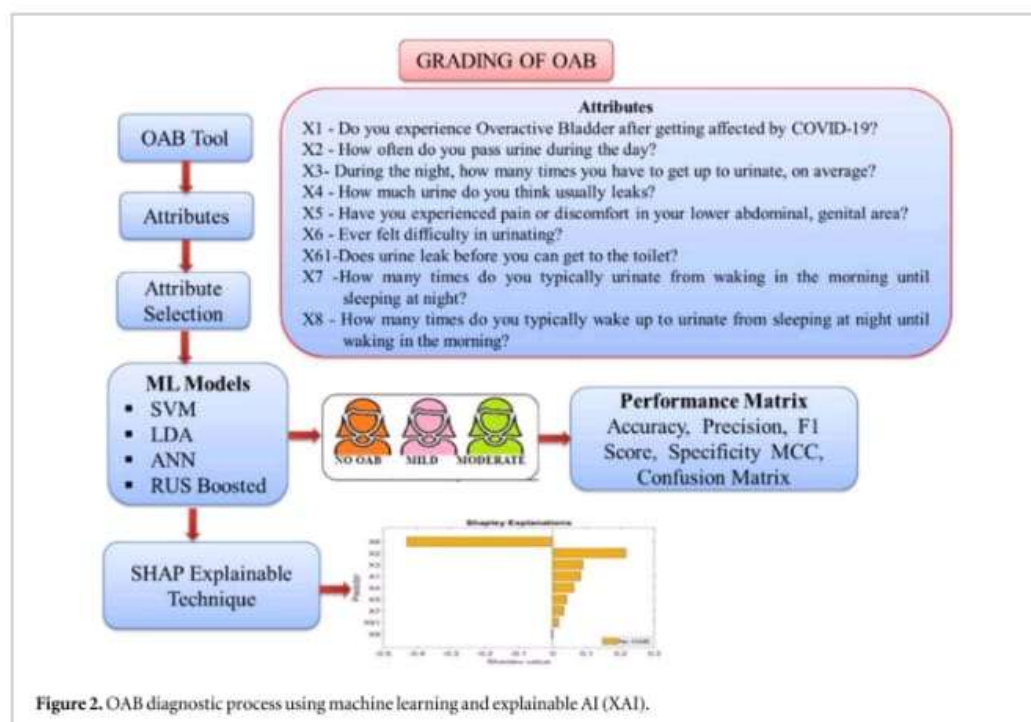


Figure 2. OAB diagnostic process using machine learning and explainable AI (XAI).

models like XGBoost, RF, SVM, and Multi-Layer Perceptron (MLP) have been combined with SHAP to identify bloodstream infections [33]. The goal is to make AI systems more transparent while maintaining their performance. This is important for classifying OAB, where understanding the model's decisions is crucial.

3. Results

3.1. Statistical analysis

To evaluate the discriminatory capability of the features across the three classes (No OAB, Mild, Moderate) and ensure their statistical significance, we conducted a one-way Analysis of Variance (ANOVA). The analysis was performed using SPSS 21.0 for Windows (SPSS Inc., Chicago, IL) at a 95% confidence level, considering $p < 0.05$ statistically significant. The results of this analysis are presented in table 2. All but one feature showed p-values below the significance threshold, indicating strong discriminatory power among the classes. The feature with a non-significant p-value was excluded from further analysis to enhance the robustness of the model.

3.1.1. Performance of the multi-classifier

K-fold cross-validation comprises the breakdown of data into 'k' subsets, also called 'folds,' for testing a machine learning model. A model's training is performed 'k' times, each with 'k-1' folds used for training and the remaining one used for validation. This ensures that any data point is not strictly used for training or validation and is free of bias or variance in the performance estimation. The final value is the mean of those model performances in the 'k' iterations. Hence, it is more reliable, robust, and representative of how well the model will likely perform when given unseen data. We performed 5-fold and 10-fold cross-validation on SVM, ANN, LDA and RUSBoost and their accuracies, recall, precision, and F1 score. MCC was obtained. In 5-fold cross-validation, the dataset is split into five subsets, and each subset is used as a validation set once, while the remaining subsets are used for training. Primarily, '5' iterations of the cross-validation process are performed, wherein the first cycle, 1-fold, is used as a testing database while the remaining 4-fold are combined to form the training database. This process is repeated for five cycles until each subpart is used as testing data.

As shown in table 3, with an accuracy of 84.44%, SVM demonstrated the highest level of performance in 5-fold classification. ANN fared somewhat better in contrast, with an accuracy of 82.22%. Both the RUSBoost, and LDA models showed reduced accuracy at 73.33%. This finding suggests that, compared with the SVM and ANN methods, the LD and RUS methods have more incredible difficulty correctly identifying features. For the SVM model, recall, F1 score and MCC values are 0.80, 0.83 and 0.93, and the ANN model has 0.83, 0.81 and 0.83. The RUS-boosted and LDA models showed reduced recall, F1 scores, and MCCs of 0.73, 0.72, and 0.75.

Table 2. p-values of the proposed attributes.

Attributes ^a	F value	p-value ^a	Attribute
X1	5.000	1.1×10^{-2}	Strong Significance
X2	3.611	3.5×10^{-2}	Strong Significance
X3	2.713	7.7×10^{-2}	Weak Significance
X4	3.017	5.0×10^{-2}	Strong Significance
X5	6.841	2.0×10^{-3}	Highly Significance
X6	15.595	1.0×10^{-5}	Extremely Strong Significance
X61	3.995	2.5×10^{-2}	Strong Significance
X7	2.830	6.9×10^{-2}	Weak Significance
X8	7.729	1.0×10^{-3}	Highly Significance

^aA p-value is a measure used in hypothesis testing to determine the strength of the evidence against the null hypothesis (H_0). It quantifies the probability of obtaining a result at least as extreme as the observed data. The significant attributes and their corresponding statistically significant p-values; $p > 5.0 \times 10^{-2}$ Not significant; $p \leq 5.0 \times 10^{-2}$ Significant; $p \leq 2.0 \times 10^{-3}$ Highly significant; $p \leq 1.0 \times 10^{-5}$ Extremely significant.

^aThe following attributes exhibit strong significance: X1 ($p = 0.011$), X2 ($p = 0.035$), X4 ($p = 0.05$), and X61 ($p = 0.025$). Additionally, X5 ($p = 0.002$) and X8 ($p = 0.001$) demonstrate high significance. In contrast, X3 and X7 show weak significance, with p-values of 0.077 and 0.069, respectively, as presented in table 2.

Table 3. 5-fold cross-validation on ML models — support vector machine, linear discriminant analysis, RUSBoost, artificial neural networks.

Parameters	SVM	ANN	LDA	RUSBoost
Precision	0.798	0.82	0.725	0.718
Recall	0.844	0.853	0.755	0.731
Accuracy	0.89	0.822	0.733	0.733
Specificity	0.827	0.908	0.851	0.843
F1score	0.93	0.828	0.721	0.72
MCC	0.844	0.861	0.744	0.75
Error rate	15.56	17.78	26.67	26.67

The findings imply that SVM is still the best model out of all those examined and that LDA and RUSBoosted might require more adjustments to increase the accuracy of their classifications.

While accuracy is often the first metric to evaluate, it can be misleading in imbalanced datasets. In the present work, we consider precision, recall, F1-score, and accuracy for model performance evaluation. Recall is the proportion of all actual positives classified correctly as positives and measures the model's ability to find all the positive instances. A high recall value indicates the model is good at identifying positives, with fewer false negatives. Low recall values show that many positive instances are missed, leading to false negatives. For precision measures of the accuracy of positive predictions, high precision when the model predicts positive is often correct. A low value indicates the model has many false positives and often over-predicts the positive class. The F1 score is a harmonic mean of precision and recall. It balances the two metrics into a single number. A high F1 Score suggests a good balance between precision and recall, and a low F1 score indicates the model has poor performance. In our study, the ANN model demonstrated a high recall value of 0.853 and a high precision value of 0.82, outperforming other models such as SVM, LDA, and RUSBoost. This indicates that ANN effectively identifies positive cases with fewer false negatives and is often correct when the model predicts a positive outcome, as shown in table 3. However, the SVM model achieved the highest F1 score of 0.93, as illustrated in table 3. Based on the F1 score, we conclude that the SVM model performs better at classifying OAB symptoms.

A confusion matrix is essential to assess a classification model's performance, representing its accuracy. It's a matrix that summarizes the predictions made by a model against the actual ground truth values. The SVM model achieves the lowest misclassification rate for all OAB classified except Moderate OAB, as shown in figure 3(A). The classifier misclassified one sample-Mild and No OAB-, but the SVM model 5-fold misclassified five moderate samples. In the ANN model, Mild was misclassified by the classifier sample 6; for Moderate and No OAB, one sample was misclassified by the classifier shown in figure 3(B). In the LDA model, mild and moderate have been misclassified by the classifier samples of 7 and 5; No OAB and all are classified perfectly classifier shown in figure 3(C), RUSBoost model mild, moderate, and No OAB have misclassified by the classifier sample of 6, 5 and 1 as shown in figure 3(D) for 5-fold cross-validation method.

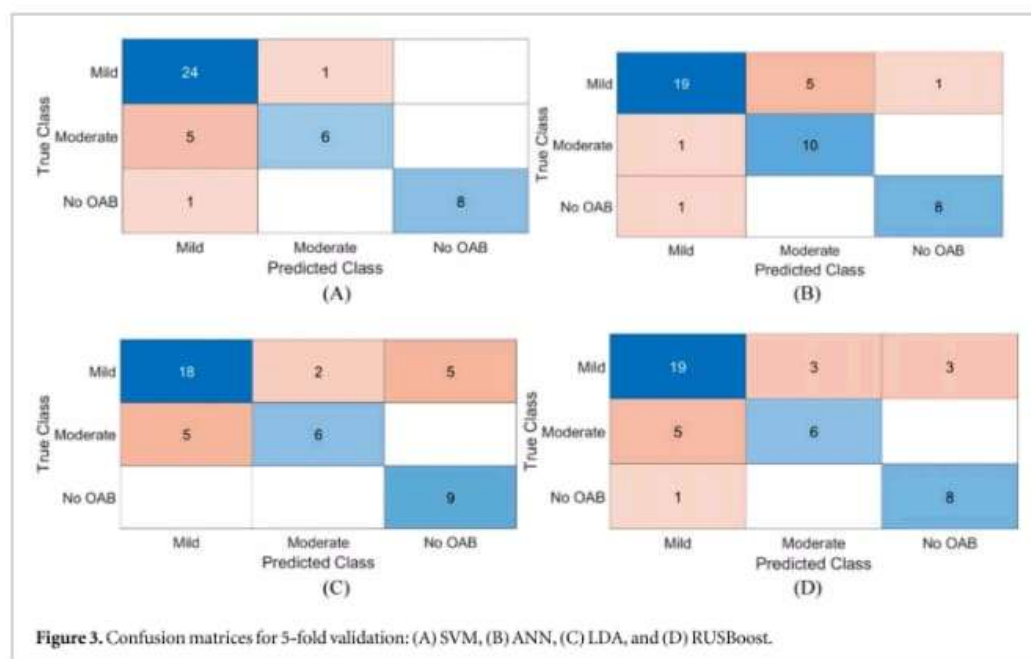


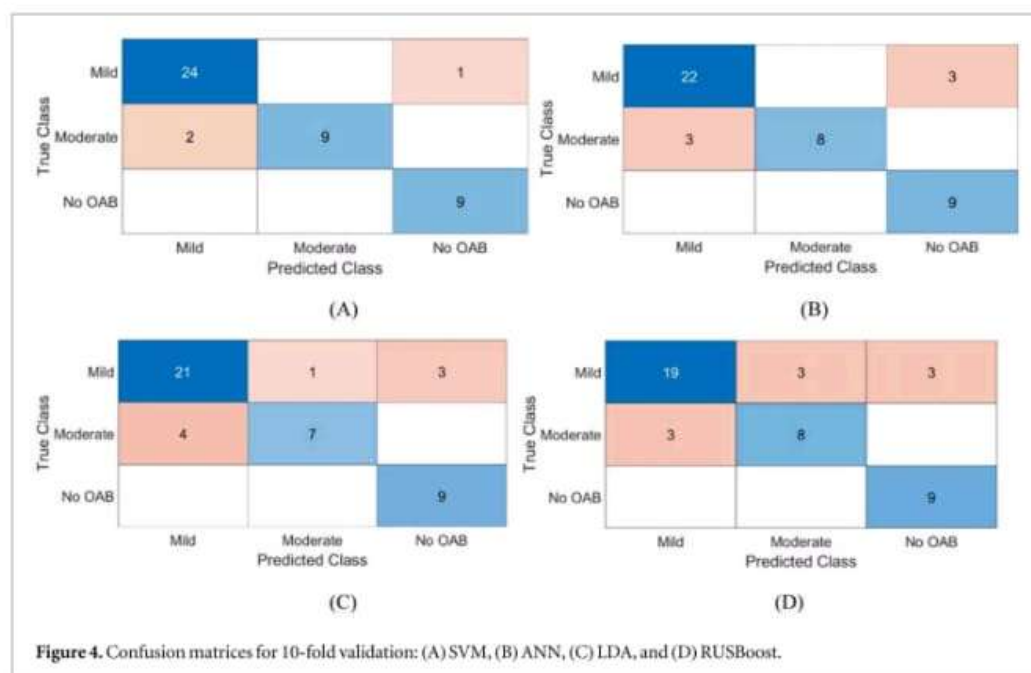
Table 4. 10-fold cross validation on ML models—support vector machine, linear discriminant analysis, RUSBoost, neural networks.

Parameters	SVM	ANN	LDA	RUSBoost
Precision	0.941	0.877	0.823	0.78
Recall	0.926	0.869	0.826	0.829
Accuracy	0.933	0.867	0.822	0.8
Specificity	0.957	0.922	0.896	0.893
F1score	0.929	0.86	0.811	0.798
MCC	0.935	0.829	0.829	0.829
Error rate	6.67	13.33	17.78	20

The classifier performance metric was improved to check if we changed 5-fold to 10-fold cross-validation. Primarily, '10' iterations of the cross-validation process are performed, wherein the first cycle, 1-fold, is used as a testing database while the remaining 9-folds are combined to form the training database. For 10-fold cross-validation, classifiers performed better with higher accuracy, precision, recall, F1 and MCC scores. For example, the SVM model achieved 0.94 precision, 0.93 recall, 0.94 MCC and 93.3% compared with ANN, LDA and RUS Boosted, as shown in table 4. Regarding the error rate, SVM is lower than ANN, LDA, and RUS Boosted, as shown in table 4. Compared to SVM, the remaining model error rate doubled.

In confusion matrices, the diagonal elements are the samples that are correctly predicted. The SVM model achieves the lowest misclassification rate for all three being classified; a total of 1 and 2 samples were misclassified by the classifier for mild and moderate, as shown in figure 4(A). Similarly, the remaining three models, ANN, LDA, and RUSBoost the misclassification rate, as shown in figures 4(B)–(D). These matrices demonstrate that 10-fold techniques significantly reduce the misclassification rate for all four models (figures 4(A)–(D)) compared to using 5-fold (figures 3(A)–(D)).

Another important metric for evaluating the classifiers is the ROC curve. An AUC value ranges from 0 to 1. A value of 0.5 suggests no discrimination (random performance), while a value closer to 1 indicates excellent model performance. AUC value: 0.9–1.0: Excellent discrimination capability. 0.8–0.9: Good discrimination capability. 0.7–0.8: Fair discrimination capability. 0.6–0.7: Poor discrimination capability. 0.5–0.6: Fail, model performs worse than random guessing. A curve that hugs the top left corner indicates a high-performing model with high TPR and low FPR. A curve along the diagonal line (from (0,0) to (1,1)) suggests a model with no discrimination capability, equivalent to random guessing. AUC values for SVM, ANN, LDA and RUSBoost using a 10-fold cross-validation method are shown in figures 5(A)–(D). The SVM model correctly distinguishes between all the positive and the negative class points of all three OABs, which have an AUC of around 0.95, as



shown in figure 5(A). Similarly, except for the moderate remaining two, ANN has an AUC above 0.9, and moderate ANN has an AUC of 0.8. It shows a moderate classifier with good discrimination capability TPR and low FPR, as shown in figure 5(B). For the LDA model, mild and moderate have AUC values of 0.8, and RUSBoost model, moderate and No OAB have AUC values above 0.9, and mild has 0.8, as shown in figures 5(C)–(D). These results demonstrate that the SVM model had Excellent discrimination capability and was most effective in classifying all three levels of OAB with AUC around 1.

The comparison between 5-fold and 10-fold cross-validation highlights a significant increase in model performance as the number of folds is increased from 5 to 10. When 10-fold cross-validation was used, the models demonstrated higher accuracy in all four models. Specifically, SVM and NN improved the accuracy by 4.6% and 5.4% 10-fold when compared to 5-fold, LDA has a 10% increase in accuracy, the error rate was reduced from 15.56 to 6.67 for 10-fold validation, precision was increased from 0.798 to 0.941 in 10-fold validation compared to 5-fold, indicating that these models are improved to the increased number of validation cycles. Even RUSBoost, which is typically more robust in handling imbalanced data, experienced an 8% increase in accuracy, as shown in figure 6.

In this case, the initial dataset contained only 53 data points, resulting in a test dataset of just 5. This is indeed a severe under-representation, which could affect the reliability of the results. To address this limitation, we employed the Synthetic Minority Over-sampling Technique (SMOTE) to augment the dataset, increasing the number of test data points to 90, thereby providing a more representative and balanced test set. This pair of settings aims to improve the models' resilience and accuracy, especially while managing underrepresented cases.

3.1.2. Performances of classifiers with synthetic minority over sampling technique (SMOTE)

The Synthetic Minority Oversampling Technique (SMOTE) is a method used to address class imbalance in classification tasks by generating synthetic samples for the minority class. It works by interpolating between existing instances of the minority class and their nearest neighbours, thereby augmenting the dataset with new synthetic examples. This process helps the model learn from underrepresented classes, improving performance metrics, reducing bias, and enhancing predictive accuracy. SMOTE helps build robust models that better generalize unseen data by providing a more balanced representation of each class. This study applied SMOTE to expand the dataset to 900 observations, with 300 observations for each category—No OAB, Mild OAB, and Moderate OAB. This significantly increased from the original dataset, which contained only 52 observations, ensuring a more equitable representation of all classes.

The analysis results show that applying SMOTE improved the accuracy of all the models. Specifically, the accuracy for the SVM increased by 2.89%, rising from 93.33% to 95.67%. The ANN model saw an 8.60% improvement, with accuracy rising from 87.71% to 95.95%. The most notable improvement was observed in the RUS Boosted model, which experienced a 13.9% increase from 80.00% to 92.94%. Similarly, the LDA model improved by 8.44%, increasing accuracy from 82.22% to 89.80%, as shown in table 5. These results highlight the

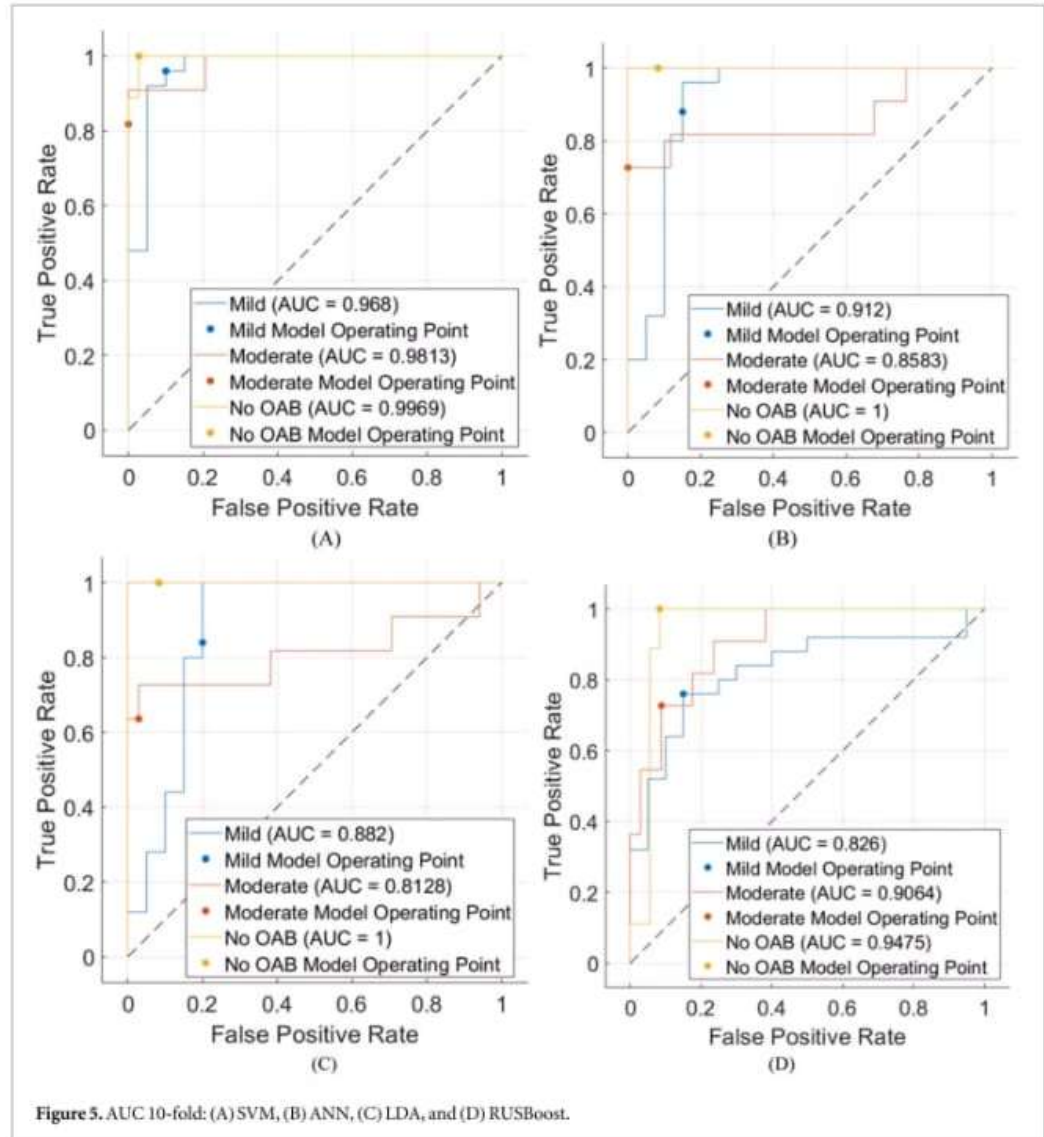


Figure 5. AUC 10-fold: (A) SVM, (B) ANN, (C) LDA, and (D) RUSBoost.

effectiveness of SMOTE in addressing class imbalance and enhancing the predictive accuracy of machine learning models.

To evaluate the inter-fold variability in terms of mean and standard deviation, we employed 10-fold cross-validation to assess the model's stability and reliability. A low standard deviation indicates that the prediction model is stable and performs consistently across different folds. Conversely, a high standard deviation suggests significant performance variation across folds, implying potential instability or dependency on specific data subsets. As shown in table 6, all models exhibited high standard deviation without applying SMOTE, indicating potential instability or sensitivity to data distribution. However, after applying SMOTE, the standard deviation decreased across all four models, with SVM demonstrating the lowest standard deviation. This suggests that the SVM model achieves higher stability when trained with SMOTE and maintains consistent performance across different folds.

3.2. Explainable AI (XAI) with SHAP -multi-class classification

The Shapley value of a feature represents its contribution to the difference between the true value and the predicted class. A positive Shapley value indicates that the feature increases the model's prediction relative to the baseline (or expected) prediction, while a negative value suggests that the feature lowers the prediction. The absolute value of the Shapley value reflects the strength of a feature's influence on the prediction, with larger absolute values indicating a more substantial impact. In this study, MATLAB R2023a was used to retrain the model on the full dataset. The SHAP summary plot was employed to identify which features most significantly impact the model's predictions and to illustrate the direction (positive or negative) and strength of their influence.

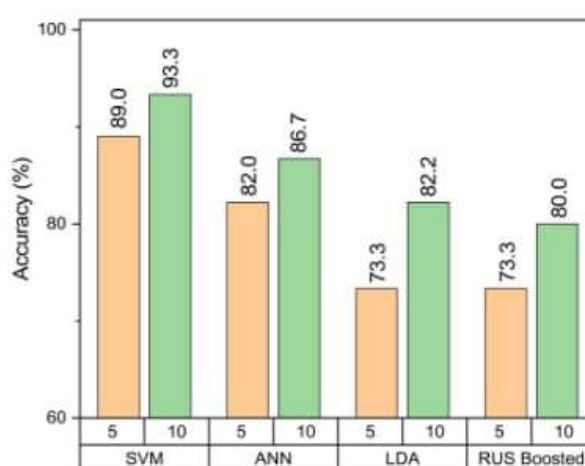


Figure 6. Accuracies of ML models—support vector machine (SVM), artificial neural network (ANN), linear discriminant analysis (LDA), and RUSBoost, in 5-fold and 10-fold cross validation.

Table 5. Comparison of the accuracies of the ML models with and without the SMOTE technique.

Model	Without SMOTE (%)	With SMOTE (%)
SVM	93.33	95.67
ANN	87.7	95.95
LDA	82.22	89.80
RUSBoost	80.0	92.94

Table 6. Comparison of the accuracies (Mean and Standard Deviation (SD)) of cross-validation performance with and without the SMOTE technique.

Model	Without SMOTE (%)		With SMOTE (%)	
	Mean	SD	Mean	SD
SVM	0.9000	0.1414	0.9656	0.0192
ANN	0.9200	0.1687	0.9378	0.0420
LDA	0.8800	0.1398	0.9078	0.0297
RUS Boosted	0.7600	0.2066	0.9322	0.0225

The attributes considered in this analysis are as follows:

X1: Do you experience Overactive Bladder (OAB) after being affected by COVID-19?

X2: How often do you pass urine during the day?

X3: On average, how many times do you wake up during the night to urinate?

X4: How much urine do you typically leak, regardless of whether you use pads or protection?

X5: Have you experienced pain or discomfort in the lower abdominal or genital area?

X6: Have you ever experienced difficulty urinating?

X61: Does urine leak before you can reach the toilet?

X7: How often do you typically urinate between waking up in the morning and sleeping at night?

X8: How often do you typically wake up to urinate from night to morning?

Figures 7(A)–(C) show the ranking of features by their Shapley value, reflecting their overall importance in predicting OAB levels. All attributes have a positive Shapley value for the ‘No OAB’ class, except for X61, X1, X2, X3, and X4, which drive the model’s prediction higher, as depicted in figure 7(A). For the ‘Mild OAB’ class, X1 and X3 have negative Shapley values, while the remaining features, particularly X5, X6, X2, and X61, contribute positively to the prediction, as shown in figure 7(B). Similarly, for the ‘Moderate OAB’ class, most attributes exhibit negative Shapley values, except X6 and X1, as seen in figure 7(C). By leveraging SHAP values, we can

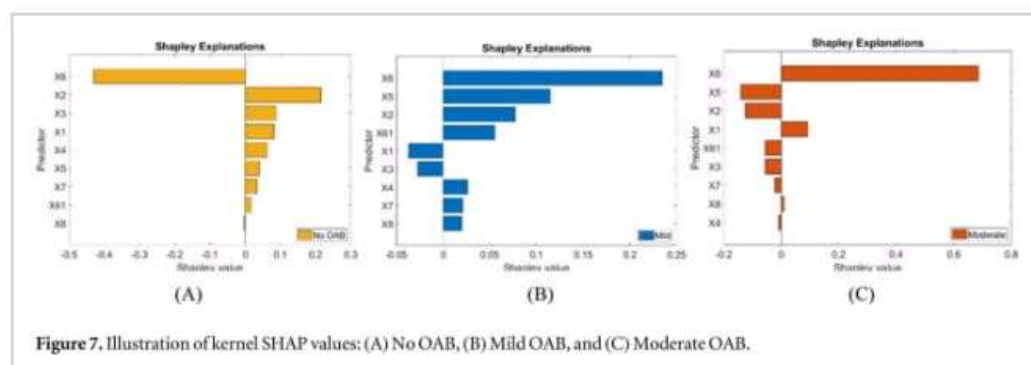


Figure 7. Illustration of kernel SHAP values: (A) No OAB, (B) Mild OAB, and (C) Moderate OAB.

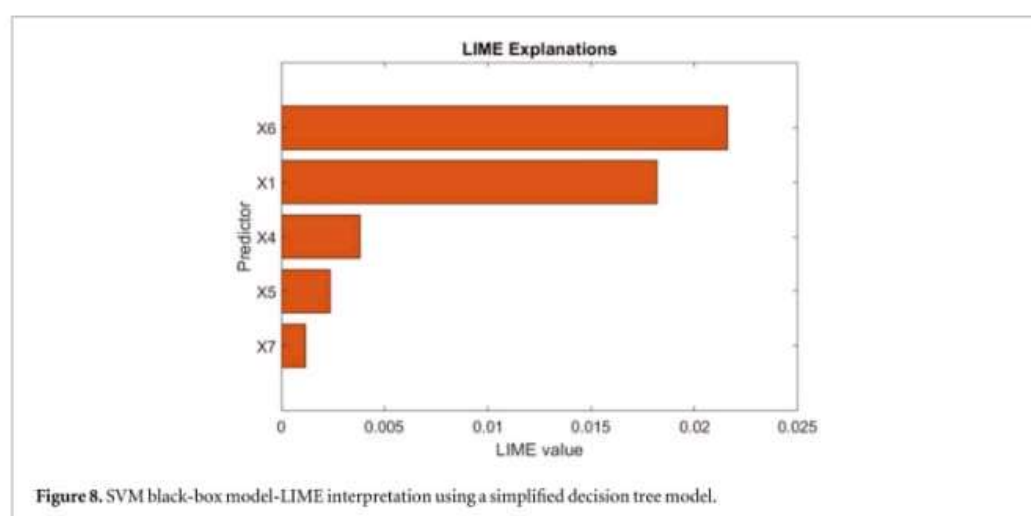


Figure 8. SVM black-box model-LIME interpretation using a simplified decision tree model.

improve trust and interpretability in complex models. This is particularly valuable in fields that demand transparency, such as healthcare and legal domains, where understanding the decision-making process of classifiers is crucial, especially in predicting OAB diagnoses.

XAI aids researchers, developers, domain experts, and end-users in understanding these models' internal workings. Several XAI techniques exist, including Local Interpretable Model-Agnostic Explanations (LIME) and SHapley Additive Explanations (SHAP), which provide insights into model decision-making. LIME facilitates the interpretation of individual predictions by approximating a complex model with a simpler, interpretable one at a specific query point. For instance, in the case of a linear model, the estimated coefficients indicate the contribution of each predictor, whereas in a decision tree, predictor importance values serve the same purpose. When applying LIME to a decision tree model, the estimated predictor importance values measure each feature's influence on the model's output. Based on their absolute importance, the bar graph representation of LIME values highlights the most influential predictors, displaying only the subset of attributes included in the simplified model.

The results indicate that attributes X6, X1, and X4 strongly influence model predictions, particularly in the context of an SVM model, where these features play a significant role, as shown in figure 8. The longer the bar in the visualization, the greater the feature's importance in determining the model's prediction. Additionally, both LIME and SHAP analyses highlight that X6, X5, and X7 contribute significantly to model predictions. By employing these explainability techniques, we enhance transparency in AI/ML models, ultimately fostering trust in their decision-making processes.

4. Discussion

Overactive bladder is a syndrome marked by storage symptoms and can stem from various underlying causes. It is primarily characterized by a sudden, intense urge to urinate, with or without involuntary urine leakage. OAB is frequently associated with increased urinary frequency during both day and night. As a multifaceted condition, OAB arises from numerous contributing factors and mechanisms [4]. Despite significant advancements in

Table 7. Comparison with state-of-the-art approaches.

Article	Clinical purpose	Algorithms used	Highest accuracy (%)
Burton <i>et al</i> [34]	Application of supervised machine learning to detect urinary tract infections	Heuristic models using microscopy thresholds, multilayer feed-forward Neural Network, Random Forest Classifier, XGBoost Gradient Boosted Tree Classifier	Classification sensitivity of 95 $\pm$ 0.2%
Laurikkala <i>et al</i> [35]	Comparison of models to diagnose female urinary incontinence	Galactica, discriminant analysis, logistic regression, k-means cluster analysis, C4.5 decision-tree generator, random bit climber hill-climbing algorithm	91% accuracy in C4.5 rule-based and 90% accuracy in Galactica
Karam <i>et al</i> [36]	Real-time bladder event detection based on an <i>in situ</i> sensor	Static Detrusor Thresholding (SDT), Hybrid Detrusor Thresholding (HDT), Context Aware Thresholding (CAT), Global Detrusor Thresholding (GDT), Global Vesical Thresholding (GVT), Adaptive Vesical Thresholding (AVT)	97% accuracy with a 1.34% false positivity rate in CAT
Huang and Chen [37]	Diagnosis of urodynamic stress incontinence using computer-aided vector-based perineal ultrasound	Multilayer Perceptron (MLP) neural network classifier, K-fold cross-validation	91.7% accuracy; 94.4% positive predictive value
Proposed Approach—ML with XAI	OAB Questionnaires—Prediction of the level of Overactive Bladder (OAB) symptoms	Support Vector Machine, Linear Discriminant Analysis, Artificial Neural Networks, RUS Boosted, K-fold cross-validation, SMOTE	93.33% accuracy in SVM (without SMOTE); 95.67% accuracy in SVM, 95.8% positive predictive value (with SMOTE)

research, the development of reliable diagnostic methods for OAB has been slowed by the considerable time and resources required to ensure their safe integration into treatment. Traditional diagnostic approaches for OAB often involve invasive biochemical tests, which require precise steps, leading to long development timelines and high costs. Recently, a growing focus has been on accelerating OAB diagnosis through questionnaires, utilizing ML techniques to improve efficiency [23]. Integrating ML with high-throughput theoretical models and subjective data is moving from labour-intensive manual processes toward more data-driven, streamlined approaches.

Support vector machine has emerged as a leading method among various ML techniques due to its ability to identify subtle patterns in complex datasets [18]. SVM has been applied effectively in diverse areas, such as handwriting recognition, fraud detection, speaker and facial identification [34], and cancer research, where it helps uncover genomic patterns associated with subtypes, prognostic outcomes, drug responses, and tumour-specific biological processes [35]. In this study, SVM outperformed other ML models, such as ANN, LDA, and RUSBoost, classifying OAB symptoms based on subjective assessments (OAB questionnaires). SVM enhanced prediction accuracy and streamlined the diagnostic process, offering a more efficient and cost-effective solution for bladder condition diagnosis.

Class imbalance, where specific categories are underrepresented in the dataset, is a common challenge in data analysis and can lead to biased predictions. Resampling methods like SMOTE are commonly used to address this issue. In our study, applying SMOTE balanced the dataset, improving the SVM model's accuracy and positively impacting the performance of other models. These findings suggest that SMOTE can enhance predictive accuracy, reduce training time, and simplify model complexity compared to imbalanced datasets. This study also integrates XAI techniques to improve the interpretability of machine learning models, aiming to support the scientific community in better identifying OAB. This approach is expected to lead to more accurate and timely interventions for predicting OAB symptoms by improving transparency. Incorporating this AI framework into clinical practice allows healthcare professionals to make more informed decisions based on AI-generated insights, enabling more precise treatment choices for patients with bladder-related conditions.

Table 7 compares state-of-the-art approaches with the proposed method, demonstrating that the proposed approach achieves superior performance based on accuracy evaluation metrics. All these techniques rely on biochemical variables or sensors to diagnose urinary incontinence. The proposed method shares a goal similar to existing approaches, as we aim to identify OAB in females. However, the proposed methodology and theoretical framework differ significantly, offering a more practical and convincing solution. Unlike previous studies, our approach utilizes self-assessment questionnaires to classify OAB symptoms, achieving an accuracy exceeding 95%. This innovation advances the understanding and applications of earlier research without relying on biomarkers or sensors. For instance, Karam *et al* [36] employed *in situ* sensors to classify real-time bladder events

with an accuracy of 97%. While effective, these methods have a key limitation: the need for additional hardware, which can impose significant costs on healthcare systems. In contrast, our proposed system eliminates the need for expensive hardware by classifying OAB symptoms using subjective questionnaire responses. This makes our approach more accessible and cost-effective, especially in resource-limited healthcare settings.

5. Conclusion

The study introduces an innovative, non-invasive diagnostic approach for overactive bladder using questionnaire-based data, advancing the field of OAB diagnosis. The study successfully classified OAB into three categories, No OAB, Mild OAB, and Moderate OAB, using machine learning models, including SVM, ANN, LDA, and RUSBoost. Rigorous cross-validation (both 5-fold and 10-fold) showed enhanced model performance, with SVM yielding the highest accuracy. The study also employed the SMOTE to address the class imbalance, improving accuracy across all models. SMOTE demonstrated its value in optimizing predictive models by increasing accuracy, mitigating bias, and simplifying model training. These results highlight the crucial role of such techniques in enhancing model reliability, which is vital for clinical diagnostics.

Moreover, SHapley Additive Explanations (SHAP) provided transparency into model behaviour, ensuring interpretability. SHAP values explained the influence of individual questionnaire responses on the model's predictions, thus fostering trust and understanding of the model's decision-making process. In healthcare, where interpretability is critical, SHAP values help bridge the gap between machine learning models and clinical adoption. In conclusion, this study emphasizes the transformative potential of machine learning, particularly when combined with SMOTE and SHAP values, in improving OAB diagnosis. These techniques offer more accurate, interpretable tools to enhance clinical decision-making significantly. This approach could be incredibly impactful in resource-limited or rural healthcare settings, where advanced diagnostic tools are often scarce. Future research could build on these findings by expanding datasets, integrating additional AI methods, and exploring deep learning techniques further to improve diagnostic capabilities for OAB and other medical conditions.

Ethical statement

The study does not involve with invasive procedures or bodily generated signals. All guidelines are followed while conducting the study according to the Institutional policies. Informed consent was obtained from all participants.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability statement

The data cannot be made publicly available upon publication because they contain sensitive personal information. The data that support the findings of this study are available upon reasonable request from the authors.

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or time too. Poor people cannot afford costly tests. People in villages have no specialist doctors nearby. So, most people just suffer without knowing what exactly is wrong. Questionnaires like OABSS are used to measure symptoms [6]. But manual reading of these forms takes too much time [7]. Machine learning is now used in many medical fields [10]. It helps doctors make better decisions faster [11]. Big data and machine learning together can improve healthcare a lot [12]. Machine learning finds hidden patterns in data that humans cannot find easily [13]. In urology also machine learning is now being used widely [14]. Various studies have shown that computer programs can predict urinary problems with good accuracy [15]. Artificial intelligence is changing the way doctors diagnose diseases [16]. For OAB specifically machine learning has shown very good results. SVM model was used to classify bladder conditions and gave good accuracy [9]. Real time classification of bladder events was done using machine learning [9]. Urinary tract infections were detected using AI with 95 percent sensitivity [8]. Cystitis was detected in ultrasound images using deep learning [19]. Bladder cancer was predicted using clinical lab data and machine learning [17]. Inflammatory process in

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